

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	7
Bill of Material:	9
MDMT Summary:	10
Warnings and Errors:	12
Input Echo:	13
XY Coordinate Calculations:	23
Internal Pressure Calculations:	24
Element: Bottom D end	25
Element: shell 1	28
Element: Shell 2	30
Element: Top D end	32
Shop Pressure Test Results:	35
External Pressure Calculations:	37
Element and Detail Weights:	38
Nozzle Flange MAWP:	41
Natural Frequency Calculation:	43
Wind Load Calculation:	44
Earthquake Load Calculation:	47
Wind/Earthquake Shear, Bending:	50
Wind/Earthquake Deflection:	51
Longitudinal Stress Constants:	52
Longitudinal Allowable Stresses:	53
Longitudinal Stresses due to:	54
Stress due to Combined Loads:	56
Skirt Stress Calculations:	60
Center of Gravity Calculation:	65
Basering Calculations:	66
Nozzle Calcs.: N3_2"	70
Nozzle Calcs.: MH_24"	76
Nozzle Calcs.: K1B_3"	82
Nozzle Calcs.: K2B_3"	87
Nozzle Calcs.: K2D_3"	92
Nozzle Calcs.: K2F_3"	97
Nozzle Calcs.: K1A_3'	102
Nozzle Calcs.: K2A_3"	107
Nozzle Calcs.: K2C_3"	112
Nozzle Calcs.: K2E_3"	117
Nozzle Calcs.: K3A_1.5"	122
Nozzle Calcs.: K3B_1.5"	125
Nozzle Calcs.: N4_2"	128
Nozzle Calcs.: N1B_10"	131
Nozzle Calcs.: N1A_10"	145
Nozzle Calcs.: N2_14"	159



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-V-32002 - 7%.pvdb

Date of Analysis : Aug 11, 2026 5:14pm

PV Elite 28 U2, April 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2100.000 mm ID	
Vessel Design Length, Tangent to Tangent	4200.00 mm
Distance of Bottom Tangent above Grade	2300.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	2000.00 mm
Internal Design Temperature	80 °C
Internal Design Pressure	2.300 MPa
External Design Temperature	80 °C
External Design Pressure	0.101 MPa
Maximum Allowable Working Pressure	2 MPa
External Max. Allowable Working Pressure	1 MPa
Shop Test Pressure	3.011 MPa
Required Minimum Design Metal Temperature	-33.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code IS 875 (Part 3):2015 Amd. 1&2 2020
 Earthquake Design Code Response Spectrum

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-516 70	...	...	K02700	+IT
Nozzle	SA-333 6	...	...	K03006	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt	...	...	...	1.6000	N/A	No
Bottom D end	2.307	0.10	2.000	1.5000	No	No
shell 1	2.304	0.10	2.000	1.5000	N/A	No
Shell 2	2.300	0.10	2.000	1.5000	N/A	No
Top D end	2.300	0.10	2.000	1.5000	No	No

Liquid Level: 1175.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.613

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
--------------	--------------	-------------------	----------------------	-----------------------	----------------------	----------------------	----------	----------

Skirt	0.0	2000.0	10.0	10.0	...	...	0.70	0.55
Ellipse	50.0	50.0	24.0	20.0	19.1	7.0	1.00	1.00
Cylinder	3050.0	3000.0	20.0	20.0	19.2	10.8	1.00	1.00
Cylinder	4150.0	1100.0	20.0	20.0	19.2	10.8	1.00	1.00
Ellipse	4200.0	50.0	24.0	20.0	19.0	7.0	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3_2"	WRC-107/537	0.855	n/a	Passed
N1B_10"	WRC-297	0.949	Shell	Passed
N1A_10"	WRC-297	0.949	Shell	Passed
N2_14"	WRC-107/537	0.852	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1000	51758.6	11303.1	235291872	57018928
20	30	2025	44050.8	10829.4	139443648	35104160
30	40	3550	43844.7	9475.17	137245376	34581176
40	50	5600	26074.8	5508.91	32323594	6175984
50	60	6175	21541.2	2273.57	6124157	113725

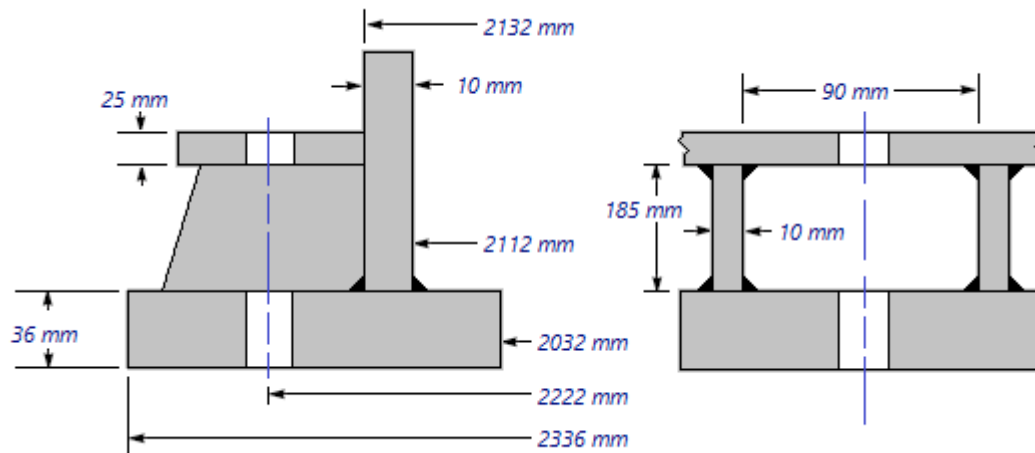
Abs Max of the all of the Stress Ratio's : 0.4673

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	36.0000	mm
Inside Diameter of Basing	2032.0000	mm
Outside Diameter of Basing	2336.0000	mm
Nominal Diameter of Bolts	24.0000	mm
Diameter of Bolt Circle	2222.0000	mm
Number of Bolts	16	

Thickness of Gusset Plates	10.0000	mm
Average Width of Gusset Plates	102.0000	mm
Height of Gussets	185.0000	mm
Distance between Gussets	90.0000	mm
Thickness of Top Plate or Ring	25.0000	mm
Radial Width of the Top Plate	119.0000	mm
Number of Gussets	32	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	51759.	N
Total Earthquake Shear on Support	11303.	N
Wind Moment on Support	239529776.	N-mm
Earthquake Moment on Support	61256836.	N-mm
Maximum Initial Bolt Load	15354.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	11909.9	kgm
Shop Test - Fabricated + water (full)	28874.2	kgm
Shipping - Fab. + rem. intls.+ shipping app.	18903.0	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	18903.0	kgm
Empty - Fab. + intls. + details + wghts.	18903.0	kgm
Operating - Empty + operating liquid (no ca)	21026.3	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3_2"	2	...	...	OK	6.30	OK	No Calc[*]
MH_24"	2	OK	...	...		OK	Passed
K1B_3"	2	OK	...	OK	7.23	OK	Passed
K2B_3"	2	OK	...	OK	7.23	OK	Passed
K2D_3"	2	OK	...	OK	7.23	OK	Passed
K2F_3"	2	OK	...	OK	7.23	OK	Passed
K1A_3'	2	OK	...	OK	7.23	OK	Passed
K2A_3"	2	OK	...	OK	7.23	OK	Passed
K2C_3"	2	OK	...	OK	7.23	OK	Passed
K2E_3"	2	OK	...	OK	7.23	OK	Passed
K3A_1.5"	2	...	...	OK	6.02	OK	No Calc[*]
K3B_1.5"	2	...	...	OK	6.02	OK	No Calc[*]
N4_2"	2	...	...	OK	6.30	OK	No Calc[*]
N1B_10"	2	OK	...	OK	9.60	OK	Passed
N1A_10"	2	OK	...	OK	9.60	OK	Passed
N2_14"	2	OK	...	OK	9.83	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2 Nozzle : MH24"
 Minimum MAWP Shells/Flanges : 2 Element : shell 1
 Minimum MAPnc Shells/Flanges : 3 Element : Shell 2

Computed Vessel M.A.W.P. : 2 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N3_2"	0.000	0.000	121.328
30	MH_24"	750.000	0.000	1225.200
30	K1B_3"	150.000	180.000	158.400
30	K2B_3"	150.000	90.000	158.400
30	K2D_3"	150.000	270.000	158.400
30	K2F_3"	150.000	125.000	158.400
30	K1A_3'	750.000	180.000	158.400
30	K2A_3"	750.000	90.000	158.400
30	K2C_3"	750.000	270.000	158.400
30	K2E_3"	750.000	125.000	158.400
30	K3A_1.5"	750.000	240.647	107.104
30	K3B_1.5"	150.000	240.647	107.104
30	N4_2"	200.000	145.000	121.328
30	N1B_10"	1855.000	0.000	499.293
30	N1A_10"	1855.000	180.000	499.293
50	N2_14"	0.000	0.000	717.200

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

PV Elite 28 U2 Licensee: SL User
FileName : 401-V-32002_ - 7%
Nozzle Summary:

Step: 42 5:15pm Aug 11, 2026

Page 6 of 168



HEXAGON

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K3A_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	244.75	300
K3B_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	244.75	300
N3_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	226.84	300
N4_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	334.27	300
K1B_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	328.72	300
K2B_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	328.72	300
K2D_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	328.72	300
K2F_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	328.72	300
K1A_3'	3.000 in	Actual	LWN	4.620	20.574	...	...	328.72	300
K2A_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	328.72	300
K2C_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	328.72	300
K2E_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	328.72	300
N1B_10"	10.000 in	80	WNF	10.750	15.088	473.05	20.00	262.84	300
N1A_10"	10.000 in	80	WNF	10.750	15.088	473.05	20.00	262.84	300
N2_14"	14.000 in	Actual	WNF	14.787	10.000	615.60	20.00	351.50	300
MH_24"	24.000 in	Actual	WNF	25.575	20.000	1219.20	16.00	238.87	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:
 Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K3A_1.5"	SA-350 LF2	16.000	10.000	...	...	...
K3B_1.5"	SA-350 LF2	16.000	10.000	...	...	...
N3_2"	SA-350 LF2	15.000	10.000	...	...	...
N4_2"	SA-350 LF2	16.000	10.000	...	...	...
K1B_3"	SA-350 LF2	16.000	10.000	...	...	...
K2B_3"	SA-350 LF2	16.000	10.000	...	...	...
K2D_3"	SA-350 LF2	16.000	10.000	...	...	...
K2F_3"	SA-350 LF2	16.000	10.000	...	...	...
K1A_3'	SA-350 LF2	16.000	10.000	...	...	...
K2A_3"	SA-350 LF2	16.000	10.000	...	...	...
K2C_3"	SA-350 LF2	16.000	10.000	...	...	...
K2E_3"	SA-350 LF2	16.000	10.000	...	...	...
N1B_10"	SA-333 6	16.000	10.000	14.000	20.000	...
N1A_10"	SA-333 6	16.000	10.000	14.000	20.000	...
N2_14"	SA-516 70	15.000	10.000	14.000	20.000	...
MH_24"	SA-516 70	16.000	10.000	12.000	16.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K3A_1.5"	750.000	240.6	207.96	0.00	shell 1
K3B_1.5"	150.000	240.6	207.96	0.00	shell 1
N3_2"	...	0.0	206.38	0.00	Bottom D end
N4_2"	200.000	145.0	313.43	0.00	shell 1
K1B_3"	150.000	180.0	307.08	0.00	shell 1



K2B_3"	150.000	90.0	307.08	0.00	shell 1
K2D_3"	150.000	270.0	307.08	0.00	shell 1
K2F_3"	150.000	125.0	307.08	0.00	shell 1
K1A_3'	750.000	180.0	307.08	0.00	shell 1
K2A_3"	750.000	90.0	307.08	0.00	shell 1
K2C_3"	750.000	270.0	307.08	0.00	shell 1
K2E_3"	750.000	125.0	307.08	0.00	shell 1
N1B_10"	1855.000	0.0	233.92	0.00	shell 1
N1A_10"	1855.000	180.0	233.92	0.00	shell 1
N2_14"	...	0.0	322.22	0.00	Top D end
MH_24"	750.000	0.0	167.38	0.00	shell 1



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 24.0mm THK X 2100.0mm ID X 50.0mm	SA-516 70
2	INSULATION: 575mm X 70mm THK	CC
1	PLATFORM: 1200mm X 1500mm WIDE	
1	INSULATION: 3000mm X 70mm THK	CC
1	INSULATION: 1100mm X 70mm THK	CC
1	PLATFORM: 1200mm X 4140mm WIDE	
1	LADDER: 6000.0mm	...
1	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-350 LF2
1	BASERING: 2336.0mm OD X 2032.0 ID X 36.0mm	SA-516 70
16	BASERING BOLTS: 24.0mm DIA	SA-36
16	BASERING BOLT NUTS: 24.0mm DIA	...
1	BASERING TOP RING: 2370.0mm OD X 2132.0 ID X 25.0mm	...
32	BASERING GUSSET PLATES: 102.0mm X 102.0mm X 185.0mm X 10.0mm	...
1	CYLINDRICAL SKIRT SEGMENT 10.0mm THK X 4023.3mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 10.0mm THK X 2611.6mm X 2000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 20.0mm THK X 3298.6mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 4000.4mm X 1100.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 2596.8mm X 1100.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Bottom D end [10]	B	-7	-10	-29	0.957	20.000	1.00	No
Bottom D end [7]	B	-2	-14	-29	0.795	24.000	1.00	No
shell 1 [8]	B	-7	-9	-29	0.965	20.000	1.00	No
Shell 2 [8]	B	-7	-9	-29	0.964	20.000	1.00	No
Top D end [10]	B	-7	-10	-29	0.954	20.000	1.00	No
Top D end [7]	B	-2	-14	-29	0.792	24.000	1.00	No
N32" [1]	B	-13	-104	-29	0.030	16.764	1.00	No
Nozzle Flg [5]	!	-46	-46					
MH24" [1]	B	-14	-29	-29	0.957	16.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
K1B3" [1]	B	-7	-29	-29	0.958	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2B3" [1]	B	-7	-29	-29	0.958	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2D3" [1]	B	-7	-29	-29	0.958	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2F3" [1]	B	-7	-29	-29	0.958	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K1A3" [1]	B	-7	-29	-29	0.957	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2A3" [1]	B	-7	-29	-29	0.957	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2C3" [1]	B	-7	-29	-29	0.957	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2E3" [1]	B	-7	-29	-29	0.957	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K3A1.5" [1]	B	-14	-104	-29	0.024	16.002	1.00	No
Nozzle Flg [5]	!	-46	-46					
K3B1.5" [1]	B	-14	-104	-29	0.024	16.002	1.00	No
Nozzle Flg [5]	!	-46	-46					
N42" [1]	B	-13	-104	-29	0.030	16.764	1.00	No
Nozzle Flg [5]	!	-46	-46					
N1B10" [1]	B	-7	-29	-29	0.957	20.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
N1A10" [1]	B	-7	-29	-29	0.957	20.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
N214" [1]	B	-7	-29	-29	0.853	20.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: -2 -9

Required Minimum Design Metal Temperature -33.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material



- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 10 30 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	2.3001	MPa
Design Internal Temperature	80.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-33.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW



Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Seismic Design Code		Spectrum
Range Type		Period
Ordinate Type		Acceleration
Include Missing Mass Components		Yes
Shock Scale Factor X Direction		1.0000
Shock Scale Factor Y Direction		0.6700
Combination Method		Group
Spectrum Name		User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.0700	0.0100
0.0120	0.0500
0.1750	0.1000
0.1750	0.1300
0.1750	0.2000
0.1750	0.2700
0.1750	0.3000
0.1750	0.4000
0.1400	0.5000
0.1270	0.5500
0.1170	0.6000
0.1050	0.6700
0.0870	0.8000
0.0700	1.0000
0.0580	1.2000
0.0500	1.4000
0.0440	1.6000
0.0390	1.8000
0.0350	2.0000
0.0290	2.4000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes



Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt
Distance "FROM" to "TO"	2000 mm
Skirt Outside Diameter	2132 mm
Diameter of Skirt at Base	2132 mm
Skirt Thickness	10 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	80 °C
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	243 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.55
Weld is pre-Heated	No
Element From Node	10
Detail Type	Weight
Detail ID	Skirt opening
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	1470.9 N
Offset from Element Centerline	0 mm

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Elliptical
Description	Bottom D end
Distance "FROM" to "TO"	50 mm
Inside Diameter	2100 mm
Element Thickness	20 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.3001 MPa
Design Temperature Internal Pressure	80 °C
Design External Pressure	0.1013 MPa
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid 20



Dist. from "FROM" Node / Offset dist -525 mm
 Height/Length of Liquid 575 mm
 Liquid Density 0.0006123 kg/cm³

Element From Node 20
 Detail Type Insulation
 Detail ID Ins: 10
 Dist. from "FROM" Node / Offset dist -525 mm
 Height/Length of Insulation 575 mm
 Thickness of Insulation 70 mm
 Density 0.40000E-04 kg/cm³

Element From Node 20
 Detail Type Nozzle
 Detail ID N32"
 Dist. from "FROM" Node / Offset dist 0 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 406.2 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 20
 Detail Type Weight
 Detail ID vortex breaker
 Dist. from "FROM" Node / Offset dist -330 mm
 Miscellaneous Weight 4.903 N
 Offset from Element Centerline 0 mm

Element# 3/5
 Element From Node 30
 Element To Node 40
 Element Type Cylinder
 Description shell 1
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 2100 mm
 Element Thickness 20 mm
 Internal Corrosion Allowance 1.5 mm
 Nominal Thickness 20 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 2.3001 MPa
 Design Temperature Internal Pressure 80 °C
 Design External Pressure 0.1013 MPa
 Design Temperature External Pressure 80 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 30
 Detail Type Platform
 Detail ID PLAT:[1 OF 1]
 Dist. from "FROM" Node / Offset dist 0 mm
 Platform Start Angle (degrees) 0.0
 Platform End Angle (degrees) 360.0
 Platform Wind Area 432000 mm²
 Platform Weight 32225 N
 Platform Railing Weight 0 N/mm
 Platform Grating Weight 0.017 kg/cm²
 Platform Width 1500 mm
 Platform Height 1200 mm
 Platform Clearance or End Offset 230 mm
 Platform Force Coefficient 1.2
 Ladder Layout Angle 0.0



Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm
Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	600	mm
Liquid Density	0.0006123	kg/cm ³
Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	70	mm
Density	0.40000E-04	kg/cm ³
Element From Node	30	
Detail Type	Nozzle	
Detail ID	MH24"	
Dist. from "FROM" Node / Offset dist	700	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	7208.2	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +IT	
Element From Node	30	
Detail Type	Nozzle	
Detail ID	K1B3"	
Dist. from "FROM" Node / Offset dist	100	mm
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	182.67	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	30	
Detail Type	Nozzle	
Detail ID	K2B3"	
Dist. from "FROM" Node / Offset dist	100	mm
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	233.56	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	30	
Detail Type	Nozzle	
Detail ID	K2D3"	
Dist. from "FROM" Node / Offset dist	100	mm
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	233.56	N
Grade of Attached Flange	GR 1.1	



Nozzle Matl SA-350 LF2 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K2F3"
Dist. from "FROM" Node / Offset dist 100 mm
Nozzle Diameter 3 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 125.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 233.56 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K1A3"
Dist. from "FROM" Node / Offset dist 700 mm
Nozzle Diameter 3 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 180.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 182.67 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K2A3"
Dist. from "FROM" Node / Offset dist 700 mm
Nozzle Diameter 3 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 233.56 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K2C3"
Dist. from "FROM" Node / Offset dist 700 mm
Nozzle Diameter 3 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 270.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 182.67 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K2E3"
Dist. from "FROM" Node / Offset dist 700 mm
Nozzle Diameter 3 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 125.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 233.56 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K3A1.5"



Dist. from "FROM" Node / Offset dist	700	mm
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	240.647	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	95.451	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K3B1.5"	
Dist. from "FROM" Node / Offset dist	100	mm
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	240.647	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	95.451	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N42"	
Dist. from "FROM" Node / Offset dist	150	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	145.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.1	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1B10"	
Dist. from "FROM" Node / Offset dist	1805	mm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	860.15	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6	+IT

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1A10"	
Dist. from "FROM" Node / Offset dist	1805	mm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	860.15	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6	+IT

Element From Node	30	
Detail Type	Weight	
Detail ID	Manway Davit	
Dist. from "FROM" Node / Offset dist	750	mm
Miscellaneous Weight	3530.2	N
Offset from Element Centerline	1200	mm



Element#	4/5	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell 2	
Distance "FROM" to "TO"	1100	mm
Inside Diameter	2100	mm
Element Thickness	20	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.3001	MPa
Design Temperature Internal Pressure	80	°C
Design External Pressure	0.1013	MPa
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1100	mm
Thickness of Insulation	70	mm
Density	0.40000E-04	kg/cm ³
Element From Node	40	
Detail Type	Weight	
Detail ID	Demister	
Dist. from "FROM" Node / Offset dist	730	mm
Miscellaneous Weight	1470.9	N
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Ladder Rung	
Dist. from "FROM" Node / Offset dist	200	mm
Miscellaneous Weight	78.448	N
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Deflector N1B	
Dist. from "FROM" Node / Offset dist	500	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	700	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Deflector N1A	
Dist. from "FROM" Node / Offset dist	500	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	700	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Clip Wt.0	
Dist. from "FROM" Node / Offset dist	100	mm
Miscellaneous Weight	6472	N
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Lifting Wt	
Dist. from "FROM" Node / Offset dist	150	mm
Miscellaneous Weight	1961.2	N



Offset from Element Centerline

0 mm

Element# 5/5
 Element From Node 50
 Element To Node 60
 Element Type Elliptical
 Description Top D end
 Distance "FROM" to "TO" 50 mm
 Inside Diameter 2100 mm
 Element Thickness 20 mm
 Internal Corrosion Allowance 1.5 mm
 Nominal Thickness 24 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 2.3001 MPa
 Design Temperature Internal Pressure 80 °C
 Design External Pressure 0.1013 MPa
 Design Temperature External Pressure 80 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Platform
 Detail ID PLAT 2
 Dist. from "FROM" Node / Offset dist 700 mm
 Platform Start Angle (degrees) 0.0
 Platform End Angle (degrees) 0.0
 Platform Wind Area 8887680 mm²
 Platform Weight 30764 N
 Platform Railing Weight 0 N/mm
 Platform Grating Weight 0.017 kg/cm²
 Platform Width 4140 mm
 Platform Height 1200 mm
 Platform Clearance or End Offset 2070 mm
 Platform Force Coefficient 1.2
 Ladder Layout Angle 90.0
 Ladder Start Elevation 0 mm
 Ladder End Elevation 6000 mm
 Unit Weight of Ladder 0.3628 N/mm
 Platform Length (top head platform) 4140 mm

Element From Node 50
 Detail Type Insulation
 Detail ID Ins: 40
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 575 mm
 Thickness of Insulation 70 mm
 Density 0.40000E-04 kg/cm³

Element From Node 50
 Detail Type Nozzle
 Detail ID N214"
 Dist. from "FROM" Node / Offset dist 0 mm
 Nozzle Diameter 14 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 1472.8 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-516 70 +IT

Element From Node 50
 Detail Type Weight
 Detail ID Pipe davit

PV Elite 28 U2 Licensee: SL User
FileName : 401-V-32002_ - 7%
Input Echo:

Step: 1 5:14pm Aug 11,2026



HEXAGON

Dist. from "FROM" Node / Offset dist	50	mm
Miscellaneous Weight	5001.1	N
Offset from Element Centerline	0	mm

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	Skirt	...	2000	...	2000
	Bottom D end	...	2050	...	50
	shell 1	...	5050	...	3000
	Shell 2	...	6150	...	1100
	Top D end	...	6200	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt	...	...	10	1.6	2132	...	1
Bottom D end	2.3072	2.3072	24	1.5	2100	138	2
shell 1	2.3037	2.3037	20	1.5	2100	138	3
Shell 2	2.3001	2.3001	20	1.5	2100	138	4
Top D end	2.3001	2.3001	24	1.5	2100	138	5

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt	...	...	No Calc	No Calc	10	No Calc	1
Bottom D end	2.30013	2.30013	2.42119	2.62353	20	19.0762	2
shell 1	2.30013	2.30013	2.39895	2.59882	20	19.2314	3
Shell 2	2.30013	2.30013	2.40255	2.59882	20	19.2034	4
Top D end	2.30013	2.30013	2.42825	2.62353	20	19.0223	5

Minimum 2 3

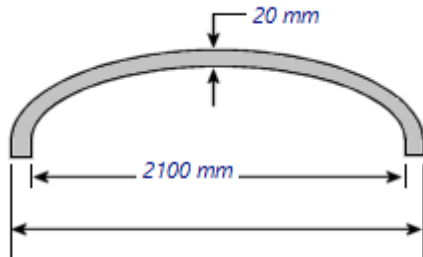
MAWP: 2 MPa, limited by: Nozzle Reinforcement.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom D end

Elliptical Head Design Information:					
Design Pressure	P	2.307 MPa	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2100.000 mm	Actual Stress		131.1 MPa
Spec. Min. Thk t,e, or sa		20.0000 mm	Spec. Nominal Thk.		24.000 mm
Surface Area		5338811.5 mm²	SG of Contents		0.613
Element Volume		1385.7 ltr	Weight of Contents		8328.2 N
Empty Weight		10457.4 N	Operating Weight		18785.6 N
			Comp. All. Ope. Stress		126.6 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.125 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.307 \cdot 2103.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 2.307)$$

$$= 17.5762 + 1.5000 = 19.0762 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 18.5) / (0.998 \cdot 2103.0 + 0.2 \cdot 18.5)$$

$$= 2 - 0.01 = 2 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 20.0) / (1.0 \cdot 2100.0 + 0.2 \cdot 20.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.307 \cdot (0.998 \cdot 2103.0 + 0.2 \cdot 18.5)) / (2 \cdot 1.0 \cdot 18.5)$$

$$= 131.118 \text{ MPa}$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2.307 \cdot 1051.5) / (138.0 \cdot 1.0 - 0.6 \cdot 2.307) + 1.5$$

$$= 19.258 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.004 MPa](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.5) / (1051.5 + 0.6 \cdot 22.5)$$

$$= 3 - 0 = 3 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (2103.0 / (2 \cdot 526.5))^2) / 6$$

$$= 0.998102$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.878 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 20.0, tr = 17.697, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.957$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -10 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 24.0, tr = 17.881, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.795$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.323 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.795	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-14 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.125 mm

EXTERNAL PRESSURE RESULTS for: Bottom D end

Elliptical Head From 20 to 30 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
18.500	2140.00	115.68	0.0012007	90.72

$$MAEP = B / (K \cdot Do / t) = 90.724 / (0.9 \cdot 115.6757) = 0.8714 \text{ MPa}$$



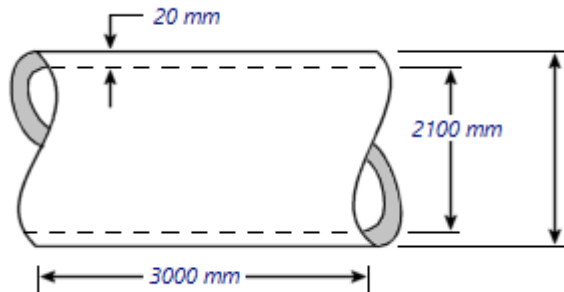
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
5.484	2140.00	390.20	0.0003559	35.59

$$MAEP = B/(K0*Do/t) = 35.5863/(0.9 * 390.2) = 0.1013 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	2.304 MPa	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2100.000 mm	Actual Stress		132.3 MPa
Spec. Min. Thk t _e , or s _a		20.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		20169026.0 mm ²	SG of Contents		0.613
Element Volume		10392.7 ltr	Weight of Contents		12492.7 N
Empty Weight		30371.1 N	Operating Weight		42863.8 N
			Comp. All. Ope. Stress		126.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.304 \cdot 1051.5) / (138.0 \cdot 1.0 - 0.6 \cdot 2.304)$$

$$= 17.7314 + 1.5000 = 19.2314 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.004 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.5) / (1051.5 + 0.6 \cdot 18.5)$$

$$= 2 - 0 = 2 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.0) / (1050.0 + 0.6 \cdot 20.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.304 \cdot (1051.5 + 0.6 \cdot 18.5)) / (1.0 \cdot 18.5)$$

$$= 132.322 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.943 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 20.0, tr = 17.854, c = 1.5 mm, E* = 1.0



Thickness Ratio = $tr * E^* / (tg - c) = 0.965$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -9 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.320 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	20.000 mm	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.965	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-9 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: shell 1

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.500	2140.00	4550.00	115.68	2.1262	0.0004883	48.82

$$MAEP = (4*B) / (3*(Do/t)) = (4*48.8187) / (3*115.6757) = 0.5627 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.301	2140.00	4550.00	230.07	2.1262	0.0001749	17.48

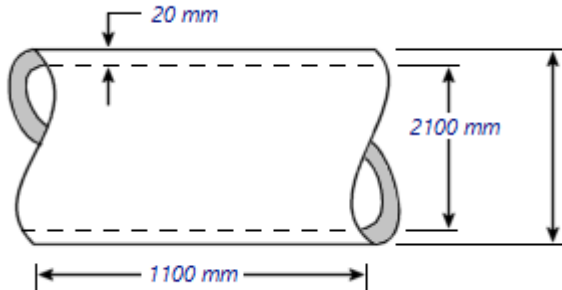
$$MAEP = (4*B) / (3*(Do/t)) = (4*17.4849) / (3*230.0741) = 0.1013 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.500	2140.00	29144.40	115.68	13.6189	0.0000881	8.80

$$MAEP = (4*B) / (3*(Do/t)) = (4*8.805) / (3*115.6757) = 0.1015 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	2.300 MPa	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		1100.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2100.000 mm	Actual Stress		132.1 MPa
Spec. Min. Thk	t,e, or sa	20.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		7395309.5 mm²	SG of Contents		0.000
Element Volume		3810.6 ltr	Weight of Contents		0.0 N
Empty Weight		11136.0 N	Operating Weight		11136.0 N
			Comp. All. Ope. Stress		126.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.3 \cdot 1051.5) / (138.0 \cdot 1.0 - 0.6 \cdot 2.3)$$

$$= 17.7034 + 1.5000 = 19.2034 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.5) / (1051.5 + 0.6 \cdot 18.5)$$

$$= 2 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.0) / (1050.0 + 0.6 \cdot 20.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.3 \cdot (1051.5 + 0.6 \cdot 18.5)) / (1.0 \cdot 18.5)$$

$$= 132.115 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.943 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 20.0, tr = 17.826, c = 1.5 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.964, Temp. Reduction = 2 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -9 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.316 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	20.000 mm	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.964	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-9 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 2

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure						MAEP
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.500	2140.00	4550.00	115.68	2.1262	0.0004883	48.82

$$MAEP = (4*B)/(3*(Do/t)) = (4*48.8187)/(3*115.6757) = 0.5627 \text{ MPa}$$

Results for Required Thickness						Tca
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.301	2140.00	4550.00	230.07	2.1262	0.0001749	17.48

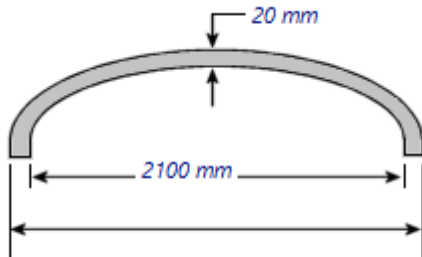
$$MAEP = (4*B)/(3*(Do/t)) = (4*17.4849)/(3*230.0741) = 0.1013 \text{ MPa}$$

Results for Maximum Stiffened Length						Slen
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.500	2140.00	29144.40	115.68	13.6189	0.0000881	8.80

$$MAEP = (4*B)/(3*(Do/t)) = (4*8.805)/(3*115.6757) = 0.1015 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top D end

Elliptical Head Design Information:					
Design Pressure	P	2.300 MPa	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2100.000 mm	Actual Stress		130.7 MPa
Spec. Min. Thk	t,e, or sa	20.0000 mm	Spec. Nominal Thk.		24.000 mm
Surface Area		5338811.5 mm²	SG of Contents		0.000
Element Volume		1385.7 ltr	Weight of Contents		0.0 N
Empty Weight		10457.4 N	Operating Weight		10457.4 N
			Comp. All. Ope. Stress		126.6 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.125 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.3 \cdot 2103.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 2.3)$$

$$= 17.5223 + 1.5000 = 19.0223 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 18.5) / (0.998 \cdot 2103.0 + 0.2 \cdot 18.5)$$

$$= 2 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 20.0) / (1.0 \cdot 2100.0 + 0.2 \cdot 20.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.3 \cdot (0.998 \cdot 2103.0 + 0.2 \cdot 18.5)) / (2 \cdot 1.0 \cdot 18.5)$$

$$= 130.716 \text{ MPa}$$

Straight Flange Required Thickness:



$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2.3 \cdot 1051.5) / (138.0 \cdot 1.0 - 0.6 \cdot 2.3) + 1.5$$

$$= 19.203 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.5) / (1051.5 + 0.6 \cdot 22.5)$$

$$= 3 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (2103.0 / (2 \cdot 526.5))^2) / 6$$

$$= 0.998102$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.878 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 20.0$, $t_r = 17.643$, $c = 1.5 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.954$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -10 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 24.0$, $t_r = 17.826$, $c = 1.5 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.792$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.316 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.792	Temperature Difference	12 °C
UCS-66(c) red.	No	Computed Min. Temp.	-14 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.125 mm

EXTERNAL PRESSURE RESULTS for: Top D end

Elliptical Head From 50 to 60 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
18.500	2140.00	115.68	0.0012007	90.72

$$MAEP = B / (K_0 \cdot Do / t) = 90.724 / (0.9 \cdot 115.6757) = 0.8714 \text{ MPa}$$

Results for Required Thickness	Tca
--------------------------------	-----

PV Elite 28 U2 Licensee: SL User
FileName : 401-V-32002_ - 7%
Element: Top D end

Page 34 of 168

Step: 7 5:14pm Aug 11,2026



HEXAGON

Tca	Outer Dia	Do/t	Factor A	Factor B
5.484	2140.00	390.20	0.0003559	35.59

$$MAEP = B / (K0 * Do / t) = 35.5863 / (0.9 * 390.2) = 0.1013 \text{ MPa}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3.011 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2.990 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3.358 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	2.548 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.289 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.289 MPa
App 27-4	= M.A.W.P.	2.316 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 2.316 • 1.0
 = 3.011 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom D end	159.4	234.0	0.681	3.03
shell 1	161.0	234.0	0.688	3.03
Shell 2	161.0	234.0	0.688	3.03
Top D end	159.4	234.0	0.681	3.03

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_2"	Nozzle	138.00	138.00	1.000
MH_24"	Nozzle	138.00	138.00	1.000
MH_24"	Pad	138.00	138.00	1.000
K1B_3"	Nozzle	138.00	138.00	1.000
K2B_3"	Nozzle	138.00	138.00	1.000
K2D_3"	Nozzle	138.00	138.00	1.000
K2F_3"	Nozzle	138.00	138.00	1.000
K1A_3'	Nozzle	138.00	138.00	1.000
K2A_3"	Nozzle	138.00	138.00	1.000
K2C_3"	Nozzle	138.00	138.00	1.000
K2E_3"	Nozzle	138.00	138.00	1.000
K3A_1.5"	Nozzle	138.00	138.00	1.000
K3B_1.5"	Nozzle	138.00	138.00	1.000
N4_2"	Nozzle	138.00	138.00	1.000
N1B_10"	Nozzle	118.00	118.00	1.000
N1B_10"	Pad	138.00	138.00	1.000
N1A_10"	Nozzle	118.00	118.00	1.000
N1A_10"	Pad	138.00	138.00	1.000
N2_14"	Nozzle	138.00	138.00	1.000
N2_14"	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom D end	138.00	138.00	1.000
shell 1	138.00	138.00	1.000
Shell 2	138.00	138.00	1.000
Top D end	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_2"	6.41	225.00	0.028
MH_24"	48.02	179.40	0.268
K1B_3"	7.43	179.40	0.041
K2B_3"	7.43	179.40	0.041
K2D_3"	7.43	179.40	0.041
K2F_3"	7.43	179.40	0.041
K1A_3'	7.43	179.40	0.041
K2A_3"	7.43	179.40	0.041
K2C_3"	7.43	179.40	0.041
K2E_3"	7.43	179.40	0.041
K3A_1.5"	5.43	179.40	0.030
K3B_1.5"	5.43	179.40	0.030
N4_2"	6.41	179.40	0.036
N1B_10"	30.14	153.40	0.196
N1A_10"	30.14	153.40	0.196
N2_14"	55.71	179.40	0.311

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	2000	...	...	No Calc	No Calc
20	30	No Calc	2140	18.5	0.0012007	90.724
30	40	4550	2140	18.5	0.0004883	48.8187
40	50	4550	2140	18.5	0.0004883	48.8187
50	60	No Calc	2140	18.5	0.0012007	90.724

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	20	6.98437	0.10133	0.87144
30	40	20	10.8014	0.10133	0.56271
40	50	20	10.8014	0.10133	0.56271
50	60	20	6.98437	0.10133	0.87144

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	2000	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	4550	29144.4	No Calc	No Calc
40	50	4550	29144.4	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	1534.21	...	1368.87	...	107.395
20	30	1066.42	1385.69	999.773	1391.39	74.6497
30	40	3097.19	10392.7	2866.93	10422.4	216.804
40	50	1135.64	3810.65	1051.21	3821.55	79.4945
50	60	1066.42	1385.69	999.773	1391.39	74.6497
Total		7899	16974.71	7286	17026.72	552

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Wght	150	...	1000	...	Skirt opening
20	Liqd	849.3	...	-262.5	...	Liquid 20
20	Insl	20.1045	...	-237.5	...	Ins: 10
20	Nozl	44.3238	...	-668.188	...	N3_2"
20	Wght	0.5	...	-330	...	vortex breaker
30	Plat	3516.26	0.18793E-04	...	...	PLAT:[1 OF 1]
30	Liqd	1273.99	...	300	...	Liquid: 20
30	Insl	58.3888	...	1500	...	Ins: 20
30	Nozl	786.541	1133.69	700	...	MH_24"
30	Nozl	19.9323	-1203.54	100	...	K1B_3"
30	Nozl	25.4854	...	100	-1203.54	K2B_3"
30	Nozl	25.4854	...	100	1203.54	K2D_3"
30	Nozl	25.4854	-690.324	100	-985.885	K2F_3"
30	Nozl	19.9323	-1203.54	700	...	K1A_3'
30	Nozl	25.4854	...	700	-1203.54	K2A_3"
30	Nozl	19.9323	...	700	1203.54	K2C_3"
30	Nozl	25.4854	-690.324	700	-985.885	K2E_3"
30	Nozl	10.4154	-565.666	700	1005.82	K3A_1.5"
30	Nozl	10.4154	-565.666	100	1005.82	K3B_1.5"
30	Nozl	11.1404	-988.486	150	-692.145	N4_2"
30	Nozl	93.8572	1166.97	1805	...	N1B_10"
30	Nozl	93.8572	-1166.97	1805	...	N1A_10"
30	Wght	360	1200	750	...	Manway Davit
40	Insl	21.4092	...	550	...	Ins: 30
40	Wght	150	...	730	...	Demister
40	Wght	8	...	200	...	Ladder Rung
40	Wght	10	700	500	...	Deflector_N1B
40	Wght	10	-700	500	...	Deflector_N1A
40	Wght	660	...	99.9997	...	Clip Wt.
40	Wght	200	...	150	...	Lifting Wt
50	Plat	3356.84	...	700	...	PLAT 2
50	Insl	20.1045	...	287.5	...	Ins: 40
50	Nozl	160.71	...	726.112	...	N2_14"
50	Wght	510	...	50	...	Pipe davit

Total Weight of Each Detail Type:

Platforms	6873.1
Liquid	2123.3
Insulation	120.0
Nozzles	1398.5
Weights	2058.5

Sum of the Detail Weights 12573.4 kgm

Weight Summation Results: (kgm)

Fabricated	Shop Test	Shipping	Erected	Empty	Operating
------------	-----------	----------	---------	-------	-----------

Main Elements	8452.9	8452.9	8452.9	8452.9	8452.9	8452.9
Nozzles	1398.5	1398.5	1398.5	1398.5	1398.5	1398.5
Wld Weights	2058.5	2058.5	2058.5	2058.5	2058.5	2058.5
Platforms	...	...	6873.1	6873.1	6873.1	6873.1
Insulation	...	...	120.0	120.0	120.0	120.0
Ope. Liquid	...	...	...	...	...	2123.3
Test Liquid	...	16964.3	...	...	...	...
Totals	11909.9	28874.2	18903.0	18903.0	18903.0	21026.3

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	11909.9 kgm
Shop Test Wt.	- Fabricated weight + water (full)	28874.2 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	18903.0 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	18903.0 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	18903.0 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	21026.3 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	8256.8 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	15856.7 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	8493.7 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	13395751	13270089
20	30	5338811	5110243
30	40	20169026	19792034
40	50	7395309	7257079
50	60	5338811	5110243

Total	51637708.0	50539684.0 mm ²
Total	51.6	50.5 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	1791.61	1791.61	1791.61	...	1791.61
20	30	1206	2055.3	2590.85	...	1206
30	40	8082.1	9716.09	18468.4	11945954	8442.1
40	50	2266.54	2274.54	6074.86	...	2274.54
50	60	5188.73	5188.73	6573.57	...	5188.73

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	18903	21026.3	35499.3
20	30	17111.4	19234.7	33707.7



30	40	15905.4	17179.4	31116.9
40	50	7463.27	7463.27	12648.4
50	60	5188.73	5188.73	6573.57

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	11945954	11945954	11945954
20	30	11945954	11945954	11945954
30	40	11945954	11945954	11945954
40	50	...	...	...
50	60	...	...	...

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Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N3_2"	4.79	5.10	80	300	GR 1.1	...	...	...	...
MH_24"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K1B_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K2B_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K2D_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K2F_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K1A_3'	4.79	5.10	80	300	GR 1.1	...	...	...	...
K2A_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K2C_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K2E_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K3A_1.5"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K3B_1.5"	4.79	5.10	80	300	GR 1.1	...	...	...	...
N4_2"	4.79	5.10	80	300	GR 1.1	...	...	...	...
N1B_10"	4.79	5.10	80	300	GR 1.1	...	...	...	...
N1A_10"	4.79	5.10	80	300	GR 1.1	2.395	4.794	4.794	4.794
N2_14"	4.79	5.10	80	300	GR 1.1	2.198	4.794	4.794	4.794
Min Rating	4.794	5.100, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N1A_10"	302.387	0.50	12500000	6700	2.316	4.794	Pass
N2_14"	389.001	0.50	24500014	9400	2.316	4.794	Pass

Evaluating Flange on Nozzle [N1A10"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(12500000) + 4 \cdot \text{abs}(6700) \cdot 302.387 \leq \pi \cdot 302.387^3 \cdot [(4.794 - 2.316) + .500 \cdot 4.794]$$

$$= 0.20811E+09 \leq 0.42365E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1A10" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.794, 0.500 \cdot 4.794 + 4.794 - 208107184.0 / (\pi \cdot 302.387^3))$$

$$= \min(4.794, 4.797)$$

$$= 4.794 \text{ MPa}$$

Evaluating Flange on Nozzle [N214"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(24500014) + 4 \cdot \text{abs}(9400) \cdot 389.001 \leq \pi \cdot 389.001^3 \cdot [(4.794 - 2.316) + .500 \cdot 4.794]$$

$$= 0.40663E+09 \leq 0.90192E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N214" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.794, 0.500 \cdot 4.794 + 4.794 - 406632576.0 / (\pi \cdot 389.001^3))$$

$$= \min(4.794, 4.993)$$

$$= 4.794 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

PV Elite 28 U2 Licensee: SL User
FileName : 401-V-32002_ - 7%
Nozzle Flange MAWP:

Step: 11 5:14pm Aug 11,2026

Page 42 of 168



HEXAGON

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 14.4907 Hz.

The Natural Frequency for the Vessel (Ope...) is 14.1526 Hz.

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (Part 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Height to Diameter ratio: 3.053

Checking the requirement for dynamic effect of wind (9.1):

Vessel Operating Natural Frequency: 14.153 Hz

Note: Vessel Natural frequency is ≥ 1 Hz.

Note: Vessel height to diameter ratio (3.053) is ≤ 5 .

Note: Dynamic wind analysis is not required.

Design Wind Speed [Vz]:

= Basic Wind Speed $\cdot k_1 \cdot k_2 \cdot k_3 \cdot k_4$ (for the 1st element)

= $50.0 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 1.15$

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= $\max(0.7 \cdot p_z, K_d \cdot K_a \cdot K_c \cdot 0.6 \cdot V_z^2)$

= $\max(0.7 \cdot 1983.503, 1.0 \cdot 0.94 \cdot 1.0 \cdot 0.6 \cdot 57.5^2)$

= 1862.9 N/m²

Element Wind Load (operating condition):

= Wind Area $\cdot C_f \cdot p_d$

= $5.1 \cdot 0.735 \cdot 1862.93$

= 7007.1 N

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	1300.00	1.0000	1.0000	1.0000	57.50	0.7351
20	2325.00	1.0000	1.0000	1.0000	57.50	0.7351
30	3850.00	1.0000	1.0000	1.0000	57.50	0.7351
40	5900.00	1.0000	1.0000	1.0000	57.50	0.7351
50	6734.18	1.0000	1.0000	1.0000	57.50	0.7351

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency

D - Average outer insulated diameter of vessel, mm

Df - Damping Factor < 0.75 Unstable, > 0.95 Stable



Dr - Average outer insulated of top half of vessel, mm
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm
 Lc - Total length of conical section(s) of vessel, mm
 tb - Uncorroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, N
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, ≤ 0.000003 (High); $0.000003 < 0.000004$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $0.0/6200.0 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[7.32/20.34^2] \cdot 10^4 = 177.001$ [Geometry Violation]

Compute the vibration possibility. If $V_p > 0.000004$ no chance. [Vp]:

= $W / (L \cdot Dr^2)$

= $199748 / (6200.0 \cdot 2233.333^2)$

= $0.64593E-05$

Since V_p is > 0.000004 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLAT:[1 OF 1]	4320000.00	2350.00	1.86	8050.76	1.20
PLAT 2	8887680.00	7150.00	1.86	16563.09	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Skirt opening	0.00	1300.00	1.86	0.00
vortex breake	0.00	1970.00	1.86	0.00
Manway Davit	0.00	3100.00	1.86	0.00
Demister	0.00	6080.00	1.86	0.00
Ladder Rung	0.00	5550.00	1.86	0.00
DeflectorN1B	0.00	5850.00	1.86	0.00
Deflector N1	0.00	5850.00	1.86	0.00
Clip Wt.	0.00	5450.00	1.86	0.00
Lifting Wt	0.00	5500.00	1.86	0.00
Pipe davit	0.00	6500.00	1.86	0.00

The Natural Frequency for the Vessel (Ope...) is 14.1526 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1300	2558.4	5116800	1.86364	7707.85
20	30	2325	2736	136800	1.86364	206.073
30	40	3850	2736	8208000	1.86364	17769.9
40	50	5900	2736	3009600	1.86364	4533.6
50	60	6734.18	2736	1458342	1.86364	21541.2



The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Input Values:

Seismic Design Code	Spectrum
Range Type	Period
Ordinate Type	Acceleration
Include Missing Mass Components	Yes
Shock Scale Factor X Direction	1.0000
Shock Scale Factor Y Direction	0.6700
Combination Method	Group
Spectrum Name	User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.0700	0.0100
0.0120	0.0500
0.1750	0.1000
0.1750	0.1300
0.1750	0.2000
0.1750	0.2700
0.1750	0.3000
0.1750	0.4000
0.1400	0.5000
0.1270	0.5500
0.1170	0.6000
0.1050	0.6700
0.0870	0.8000
0.0700	1.0000
0.0580	1.2000
0.0500	1.4000
0.0440	1.6000
0.0390	1.8000
0.0350	2.0000
0.0290	2.4000

Computed Natural Frequencies (OPE): 4

Mode	Freq (Hz)	Freq (Rad/Sec)	Period (Sec)
1	14.6867	92.2794	0.0681
2	70.6645	443.9981	0.0142
3	75.1936	472.4552	0.0133
4	169.3481	1064.0456	0.0059

Mass Participation Factors: 4

Mode	X	Y
1	0.03060002	0.00000000
2	0.00060306	0.00000000
3	-0.00000000	0.00078838
4	0.00000737	0.00000000

Computed EigenVectors, Mass Normalized: 4

Node	Dx	Dy	Rz
Mode # 1 Freq (HZ) = 14.6867			
10	0.000000	0.000000	-0.000000
20	0.030535	0.000000	-0.000436
30	0.031542	0.000000	-0.000440
40	0.101673	-0.000000	-0.000584
50	0.128798	-0.000000	-0.000592
60	0.129993	-0.000000	-0.000592
Mode # 2 Freq (HZ) = 70.6645			
10	0.000001	0.000000	-0.000000
20	0.143062	0.000000	0.000339
30	0.143625	0.000000	0.000358



40	0.015797	-0.000000	0.001164
50	-0.052181	-0.000000	0.001241
60	-0.054919	-0.000000	0.001241

Mode # 3 Freq (HZ) = 75.1936

10	-0.000000	0.000002	0.000000
20	-0.000000	0.066632	-0.000000
30	0.000000	0.067410	-0.000000
40	-0.000000	0.103296	-0.000000
50	0.000000	0.110267	-0.000000
60	-0.000000	0.110397	-0.000000

Mode # 4 Freq (HZ) = 169.3481

10	0.000000	-0.000000	-0.000000
20	0.040752	-0.000000	0.000428
30	0.039729	-0.000000	0.000439
40	-0.136449	0.000000	-0.001228
50	0.087331	-0.000000	-0.001973
60	0.094123	-0.000000	-0.001974

Mass Percentages:

Included(X)	Included(Y)	Added(X)	Added(Y)
96.85	92.70	3.15	7.30

NOTE: In the following reports, "Max Contrib" displays the contribution of the Mode/Load combination having the maximum impact on the total (and names that Mode/Load combination).

Restraint Loads:

Node	Fx(N)	Fy(N)	Mz(N-mm)
10	11311.8	8187.6	57018927.2
Max	11013.4	8158.5	56989248.5
Contrib	1 (X)	3 (Y)	1 (X)

Element Forces and Moments:

Node	Fx(N)	Fy(N)	Mz(N-mm)
10	11303.1	8168.0	57018927.2
Max	11013.4	8158.5	56989248.5
Contrib	1 (X)	3 (Y)	1 (X)
20	11303.1	8168.0	35104159.3
Max	11013.4	8158.5	34953541.5
Contrib	1 (X)	3 (Y)	1 (X)
20	10829.4	7661.9	35104159.3
Max	10675.3	7660.3	34953541.5
Contrib	1 (X)	3 (Y)	1 (X)
30	10829.4	7661.9	34581176.7
Max	10675.3	7660.3	34419559.3
Contrib	1 (X)	3 (Y)	1 (X)
30	9475.2	5922.5	34581176.7
Max	9446.7	5887.0	34419559.3
Contrib	1 (X)	3 (Y)	1 (X)
40	9475.2	5922.5	6175984.2
Max	9446.7	5887.0	6067984.5
Contrib	1 (X)	3 (Y)	1 (X)
40	5508.9	3149.5	6175984.2
Max	5412.7	3119.1	6067984.5
Contrib	1 (X)	3 (Y)	1 (X)
50	5508.9	3149.5	113724.9
Max	5412.7	3119.1	111640.5
Contrib	1 (X)	3 (Y)	1 (X)
50	2273.6	1292.7	113724.9



Max	2231.9	1280.1	111640.5
Contrib	1 (X)	3 (Y)	1 (X)
60	2273.6	1292.7	0.0
Max	2231.9	1280.1	0.0
Contrib	1 (X)	3 (Y)	

Displacements:

Node	Dx(mm)	Dy(mm)	Rz(Deg)
10	0.0000	0.0000	0.0000
Max	0.0000	0.0000	0.0000
Contrib	1 (X)	3 (Y)	1 (X)
20	0.0238	0.0013	0.0008
Max	0.0237	0.0013	0.0008
Contrib	1 (X)	3 (Y)	1 (X)
30	0.0246	0.0014	0.0008
Max	0.0245	0.0013	0.0008
Contrib	1 (X)	3 (Y)	1 (X)
40	0.0790	0.0021	0.0010
Max	0.0790	0.0021	0.0010
Contrib	1 (X)	3 (Y)	1 (X)
50	0.1001	0.0022	0.0010
Max	0.1001	0.0022	0.0010
Contrib	1 (X)	3 (Y)	1 (X)
60	0.1010	0.0022	0.0010
Max	0.1010	0.0022	0.0010
Contrib	1 (X)	3 (Y)	1 (X)



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1000	51758.6	11303.1	235291872	57018928
20	30	2025	44050.8	10829.4	139443648	35104160
30	40	3550	43844.7	9475.17	137245376	34581176
40	50	5600	26074.8	5508.91	32323594	6175984
50	60	6175	21541.2	2273.57	6124157	113725

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	51758.6	0.01586	0.058706	0.54064E-04
20	30	44050.8	0.060061	0.061422	0.54564E-04
30	40	43844.7	0.15274	0.25694	0.71987E-04
40	50	26074.8	0.29683	0.33714	0.73483E-04
50	60	21541.2	0.33898	0.34081	0.73483E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	649.987	4062.42
20	30	695.108	4344.42
30	40	695.108	4344.42
40	50	695.108	4344.42
50	60	695.108	4344.42

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 31.000 Actual deflection : 0.341 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	66664.4	55997.4	35200424	29589920
20	30	133204	123301	69944496	64835200
30	40	133204	123301	69944496	64835200
40	50	133204	123301	69944496	64835200
50	60	133204	123301	69944496	64835200



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	91.0784	154.437	-101.828	-108.054
20	30	165.597	280.795	-126.559	-128.535
30	40	165.597	280.795	-126.559	-128.535
40	50	165.597	280.795	-126.559	-128.535
50	60	165.597	280.795	-126.559	-128.535

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	64.9073	-2.95578	78.4286
30	40	64.9073	-2.95578	78.4286
40	50	64.9073	-2.95578	78.4286
50	60	64.9073	-2.95578	78.4286

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-3.31048	-3.68233	...	0.40359	0.40359
20	30	-1.36097	-1.36097	...	0.18419	0.18419
30	40	-1.26505	-1.26505	...	0.18419	0.18419
40	50	-0.5936	-0.5936	...	...	...
50	60	-0.41269	-0.41269	...	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	...	7.94921	1.92635	...	...
20	30	...	2.15005	0.54126	...	...
30	40	...	2.11616	0.5332	...	...
40	50	...	0.49839	0.095226	...	...
50	60	...	0.094427	0.0017535	...	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	1.59806
20	30	...	...	...	0.45519
30	40	...	...	...	0.44861
40	50	...	...	...	0.084342
50	60	...	...	...	0.0015544

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	-0.14588
20	30	...	-0.062145
30	40	...	-0.048037



40	50	...	-0.025546
50	60	...	-0.010485

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.04	91.08	-11.66	101.83	0.0554	0.1145
20	0.97	165.60	-3.70	126.56	0.0059	0.0292
30	1.04	165.60	-3.57	126.56	0.0063	0.0282
40		165.60	-1.09	126.56		0.0086
50		165.60	-0.51	126.56		0.0040

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-5.46	101.83		0.0536
20		165.60	-2.06	126.56		0.0163
30		165.60	-1.95	126.56		0.0154
40		165.60	-0.70	126.56		0.0056
50		165.60	-0.42	126.56		0.0034

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.67	91.08	-12.04	101.83	0.0513	0.1182
20	0.97	165.60	-3.70	126.56	0.0059	0.0292
30	1.04	165.60	-3.57	126.56	0.0063	0.0282
40		165.60	-1.09	126.56		0.0086



50 165.60 -0.51 126.56 0.0040

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-6.16	101.83		0.0605
20		165.60	-2.15	126.56		0.0170
30		165.60	-2.03	126.56		0.0160
40		165.60	-0.71	126.56		0.0056
50		165.60	-0.42	126.56		0.0034

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	154.44	0.00	108.05	0.0000	-0.0000
20	0.00	280.79	0.00	128.53	0.0000	-0.0000
30	0.00	280.79	0.00	128.53	0.0000	-0.0000
40	0.00	280.79	0.00	128.53	0.0000	-0.0000
50	0.00	280.79	0.00	128.53	0.0000	-0.0000

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	154.44	0.00	108.05	0.0000	-0.0000
20	0.00	280.79	0.00	128.53	0.0000	-0.0000
30	0.00	280.79	0.00	128.53	0.0000	-0.0000
40	0.00	280.79	0.00	128.53	0.0000	-0.0000
50	0.00	280.79	0.00	128.53	0.0000	-0.0000

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.67	91.08	-12.04	101.83	0.0513	0.1182
20	65.88	165.60		126.56	0.3978	
30	65.94	165.60		126.56	0.3982	
40	64.81	165.60		126.56	0.3914	
50	64.59	165.60		126.56	0.3900	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-6.16	101.83		0.0605
20	64.21	165.60		126.56	0.3877	
30	64.31	165.60		126.56	0.3884	
40	64.38	165.60		126.56	0.3888	
50	64.49	165.60		126.56	0.3894	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.67	91.08	-12.04	101.83	0.0513	0.1182
20		165.60	-6.65	126.56		0.0526
30		165.60	-6.52	126.56		0.0515
40		165.60	-4.05	126.56		0.0320
50		165.60	-3.46	126.56		0.0274

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-6.16	101.83		0.0605
20		165.60	-5.10	126.56		0.0403
30		165.60	-4.99	126.56		0.0394
40		165.60	-3.67	126.56		0.0290



50 165.60 -3.38 126.56 0.0267

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	154.44	0.00	108.05	0.0000	-0.0000
20	78.43	280.79		128.53	0.2793	
30	78.43	280.79		128.53	0.2793	
40	78.43	280.79		128.53	0.2793	
50	78.43	280.79		128.53	0.2793	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	154.44	0.00	108.05	0.0000	-0.0000
20	78.43	280.79		128.53	0.2793	
30	78.43	280.79		128.53	0.2793	
40	78.43	280.79		128.53	0.2793	
50	78.43	280.79		128.53	0.2793	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-3.71	101.83		0.0365
20	63.73	165.60		126.56	0.3849	
30	63.83	165.60		126.56	0.3854	
40	64.31	165.60		126.56	0.3884	
50	64.49	165.60		126.56	0.3895	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-3.09	101.83		0.0304
20	63.65	165.60		126.56	0.3844	
30	63.74	165.60		126.56	0.3849	
40	64.36	165.60		126.56	0.3886	
50	64.53	165.60		126.56	0.3897	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-4.09	101.83		0.0401
20	63.73	165.60		126.56	0.3849	
30	63.83	165.60		126.56	0.3854	
40	64.31	165.60		126.56	0.3884	
50	64.49	165.60		126.56	0.3895	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-3.71	101.83		0.0365
20	63.73	165.60		126.56	0.3849	
30	63.83	165.60		126.56	0.3854	
40	64.31	165.60		126.56	0.3884	
50	64.49	165.60		126.56	0.3895	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-4.09	101.83		0.0401
20		165.60	-1.55	126.56		0.0122
30		165.60	-1.45	126.56		0.0115
40		165.60	-0.59	126.56		0.0047



50 165.60 -0.41 126.56 0.0033

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		75.90	-4.23	84.86		0.0499
20	63.67	138.00		105.47	0.4614	
30	63.78	138.00		105.47	0.4622	
40	64.29	138.00		105.47	0.4659	
50	64.48	138.00		105.47	0.4673	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		75.90	-4.23	84.86		0.0499
20		138.00	-4.56	105.47		0.0433
30		138.00	-4.45	105.47		0.0422
40		138.00	-3.57	105.47		0.0339
50		138.00	-3.38	105.47		0.0320

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-4.09	101.83		0.0401
20	63.73	165.60		126.56	0.3849	
30	63.83	165.60		126.56	0.3854	
40	64.31	165.60		126.56	0.3884	
50	64.49	165.60		126.56	0.3895	

Absolute Maximum of the all of the Stress Ratio's 0.4673

Governing Element: Top D end
 Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt	...	-3.8	0.4	...	...	+ Tens
Skirt	...	-3.8	-0.4	...	-4.2	- Comp
Bottom D end	64.9	-1.4	0.2	...	63.7	+ Tens
Bottom D end	64.9	-1.4	-0.2	...	...	- Comp
shell 1	64.9	-1.3	0.2	...	63.8	+ Tens
shell 1	64.9	-1.3	-0.2	...	...	- Comp
Shell 2	64.9	-0.6	...	...	64.3	+ Tens
Shell 2	64.9	-0.6	...	...	...	- Comp
Top D end	64.9	-0.4	...	...	64.5	+ Tens
Top D end	64.9	-0.4	...	...	...	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt

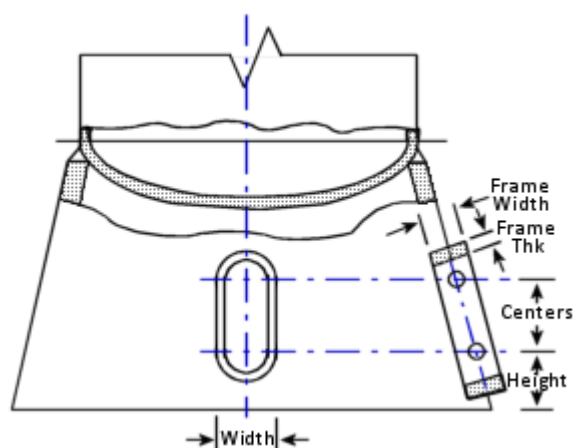
Total vessel (empty) weight	Fg	170318.031 N
Head Weight below section 2-2	DeltaFg	11826.054 N
Weight of vessel contents (liquid)	Ff	20820.961 N
Vessel Gross Weight	Fg + Ff	191139.000 N
Skirt Mean Diameter	Dz	2122.000 mm
Skirt Analysis Thickness (Corroded)	ez	8.400 mm
Skirt Height	H	2000.000 mm
Additional Axial Force at section 1-1	F1	0.000 N
Moment at section 1-1	M1	139443648.000 N-mm
Applied Moment - Bottom of Skirt	Mb	235291872.000 N-mm
Weight at Bottom of Skirt	W	206183.500 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	2121.500 mm
Head Corroded Thickness	eb	18.500 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 526.5 \text{ mm} \\
 K &= D_i / (2 * h_i) = 2103.0 / (2 * 526.5) = 1.9972 \\
 R &= D_i (0.44 * K + 0.02) = 2103.0 * (0.44 * 1.997 + 0.02) = 1890.0638 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 2103.0 * ((0.5 / 1.997) - 0.08) = 358.26 \text{ mm}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	31519186944.000 mm ⁴
Moment of Inertia	Iyy =	31519186944.000 mm ⁴
Section Modulus	W =	29707080.000 mm ³
Section Metal Area	A =	55998.270 mm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(0.31519E+11, 0.31519E+11) / 55998.27)} \\
 &= 750.24 \text{ mm}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:



$$\begin{aligned} &= r_g / L \\ &= 750.24 / 2000.0 \\ &= 0.375 \end{aligned}$$

Correct the Moment [M4]:

top of skirt is the critical end
Controlling Moment [M4] = 0.13944E+09 N-mm

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$\begin{aligned} &= -M / Z - W / A \\ &= -0.13944E+09 / 11883E+09 - 206183.5 / 55998.26953 \\ &= -4.855 \text{ MPa} \end{aligned}$$

Upwind Side:

$$\begin{aligned} &= +(M) / Z - W / A \\ &= +.13944E+09 / 11883E+09 - 206183.5 / 55998.26953 \\ &= -2.509 \text{ MPa} \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F_1 - F_g - F_f + 4 \cdot M_1 / D_z \\ &= 0.0 - 170318.031 - 20820.961 + 4 \cdot 0.13944E+09 / 2122.0 \\ &= 71607.7 \text{ N} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F_1 - F_g - F_f - 4 \cdot M_1 / D_z \\ &= 0.0 - 170318.031 - 20820.961 - 4 \cdot 0.13944E+09 / 2122.0 \\ &= -453885.7 \text{ N} \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned} &= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (71607.688 + 11826.054 + 20820.961) / (\pi \cdot 2121.5 \cdot 18.5) + 2.3 \cdot 2121.5 / (4 \cdot 18.5) \\ &= 66.79 \text{ MPa} \end{aligned}$$

Stress [σ_{1qm}]:

$$\begin{aligned} &= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (-0.5E+06 + 11826.054 + 20820.961) / (\pi \cdot 2121.5 \cdot 18.5) + 2.3 \cdot 2121.5 / (4 \cdot 18.5) \\ &= 62.53 \text{ MPa} \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$\begin{aligned} &= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (20820.961 + 11826.054) / (\pi \cdot 2121.5 \cdot 18.5) + 2.3 \cdot 2121.5 / (4 \cdot 18.5) \\ &= 66.21 \text{ MPa} \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$\begin{aligned} &= F_{zp} / (\pi \cdot D_z \cdot e_z) \\ &= 71607.688 / (\pi \cdot 2122.0 \cdot 8.4) \\ &= 1.28 \text{ MPa} \end{aligned}$$

Stress [σ_{3qm}]:

$$\begin{aligned} &= F_{zq} / (\pi \cdot D_z \cdot e_z) \\ &= -453885.688 / (\pi \cdot 2122.0 \cdot 8.4) \\ &= -8.11 \text{ MPa} \end{aligned}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$\begin{aligned} &= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b) \\ &= 1 - 0.5(2121.5 + 18.5 - 2122.0 + 8.4) / (358.26 + 18.5) \\ &= 0.965 \end{aligned}$$



$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.965) = 15.211 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(18.5^2 + 8.4^2 + 2 \cdot 18.5 \cdot 8.4 \cdot 0.965)}$$

$$= 13.348 \text{ mm}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 13.34 \cdot 71607.7 = 956236.88 \text{ N-mm}$$

$$M_q = a \cdot F_{zq} = 13.34 \cdot -453885.7 = -6.0611 \times 10^6 \text{ N-mm}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2$$

$$= 0.63 - 0.057 \left(\frac{18.5}{8.4} \right)^2$$

$$= 0.354$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.354 \cdot 6 \cdot 956237 / (\pi \cdot 2121.5 \cdot 18.5^2)$$

$$= 0.89 \text{ MPa}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.354 \cdot 6 \cdot -6061112.0 / (\pi \cdot 2121.5 \cdot 18.5^2)$$

$$= -5.63 \text{ MPa}$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.354 \cdot 6 \cdot 956237 / (\pi \cdot 2122.0 \cdot 8.4^2)$$

$$= 4.31 \text{ MPa}$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.354 \cdot 6 \cdot -6061112.0 / (\pi \cdot 2122.0 \cdot 8.4^2)$$

$$= -27.32 \text{ MPa}$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.728/83.524 = 0.0087$$

$$y = 125 \cdot eb/D_b = 125 \cdot 18.5/2121.5 = 1.09$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 1.09 = 3.982$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b \cdot ((y / y_A) \cdot \alpha - 1) / (4 \cdot eb)$$

$$= (2.3 + 0.0) \cdot 2121.5 \cdot ((15.21/40.0) \cdot 3.982 - 1) / (4 \cdot 18.5)$$

$$= 33.91 \text{ MPa}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 66.788 - 0.889 + 33.913$$

$$= 99.81 \text{ MPa}$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 66.788 + 0.889 - 33.913$$

$$= 33.76 \text{ MPa}$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 62.526 - 5.634 + 33.913$$



= 102.07 MPa

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 62.526 + -5.634 - 33.913$$

$$= 22.98 \text{ MPa}$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 66.207 + 0.889 + 33.913$$

$$= 101.01 \text{ MPa}$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 66.207 - 0.889 - 33.913$$

$$= 31.41 \text{ MPa}$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 66.207 + -5.634 + 33.913$$

$$= 94.49 \text{ MPa}$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 66.207 - -5.634 - 33.913$$

$$= 37.93 \text{ MPa}$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= 1.279 - 4.311$$

$$= -3.03 \text{ MPa}$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= 1.279 + 4.311$$

$$= 5.59 \text{ MPa}$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -8.106 - -27.323$$

$$= 19.22 \text{ MPa}$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -8.106 + -27.323$$

$$= -35.43 \text{ MPa}$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 137.997(3 - (1 / 1.5)) \cdot (66.788 / 137.997)^2$$

$$= 392.44 \text{ MPa}$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 137.997(3 - (1 / 1.5)) \cdot (62.526 / 137.997)^2$$

$$= 395.11 \text{ MPa}$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 137.997(3 - (1 / 1.5)) \cdot (66.207 / 137.997)^2$$

$$= 392.82 \text{ MPa}$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$

$$= 137.997(3 - (1 / 1.5)) \cdot (66.207 / 137.997)^2$$

$$= 392.82 \text{ MPa}$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:



$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2) \\ &= 137.997(3 - (1/1.5) \cdot (1.279/137.997)^2) \\ &= 413.98 \text{ MPa} \end{aligned}$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2) \\ &= 137.997(3 - (1/1.5) \cdot (-8.106/137.997)^2) \\ &= 413.68 \text{ MPa} \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	99.81	392.44	Pass	46
Stot1po	33.76	392.44	Pass	47
Stot1qi	102.07	395.11	Pass	48
Stot1qo	22.98	395.11	Pass	49
Stot2pi	101.01	392.82	Pass	50
Stot2po	31.41	392.82	Pass	51
Stot2qi	94.49	392.82	Pass	52
Stot2qo	37.93	392.82	Pass	53
Stot3pi	-3.03	413.98	Pass	54
Stot3po	5.59	413.98	Pass	55
Stot3qi	19.22	413.68	Pass	56
Stot3qo	-35.43	413.68	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	4394.333 mm
Center of Gravity of the Liquid	2105.004 mm
Center of Gravity of the Insulation	4100.000 mm
Center of Gravity of the Nozzles	3277.330 mm
Center of Gravity of the Added Weights (Operating)	4750.952 mm
Center of Gravity of the Added Weights (Empty)	4750.952 mm
Center of Gravity of Bare Shell New and Cold	3497.958 mm
Center of Gravity of Bare Shell Corroded	3517.627 mm
Vessel CG in the Operating Condition	3777.760 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	3947.827 mm
Vessel CG in the Test Condition	4019.801 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	2132.0000	mm
Skirt Thickness	STHK	10.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	36.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	MPa
Basing Yield Stress		262.00	MPa
Inside Diameter of Basing	DI, D4	2032.0000	mm
Outside Diameter of Basing	DOU, D5	2336.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	24.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Root Area of a Single Bolt	Area	338.0000	mm ²
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	114.00	MPa
Number of Bolts	RN, nb	16	
Diameter of Bolt Circle	DC, DBC	2222.0000	mm
Bolt Allowable Shear Stress		107.873	MPa
Thickness of Gusset Plates	TGA, e7	10.0000	mm
Width of Gussets at Top Plate	TWDT	102.0000	mm
Width of Gussets at Base Plate	BWDT	102.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	262.0	MPa
Height of Chair	h1	246.0000	mm
Distance between Gussets	RG, b6	90.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	45.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	25.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	119.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	51.0000	mm
External Corrosion Allowance	CA	1.5000	mm
Dead Weight of Vessel	DW	185362.5	N
Operating Weight of Vessel	ROW	206183.5	N
Earthquake Moment on Basing	EQMOM	61256836.0	N-mm
Wind Moment on Basing	WIMOM	239529776.0	N-mm
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Basing Analysis : Analyze Option

Basing Thickness Calculation method used : Simplified (Steel on Steel)

Calculation of Bolt Loads for Multiple Loading Conditions:

Bolt Load, Earthquake Operating Case [FB,e]:

$$= ((4 \cdot M/DC) - W)/nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 61256836/2222.0) - 206184)/16$$

$$= -5997.2 \text{ N}$$

Bolt Load, Windload Empty Case [FB,w]:

$$= ((4 \cdot M/DC) - W)/nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.23953E+09/2222.0) - 185363)/16$$

$$= 15353.7 \text{ N}$$



Bolt Load, Windload Vortex Shedding Case [FB,v]:
 $= ((4 \cdot M/DC) - W)/nb$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 0/2222.0) - 206184)/16$
 $= -12886.5 \text{ N}$

Calculation of Maximum Load per Bolt [W/Bolt], Wind + Dead Weight Condition:
 where: W = DW M = WIMOM

$= ((4 \cdot M/DC) - W)/nb$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 0.23953E+09/2222.0) - 185363)/16$
 $= 15353.7 \text{ N}$

Required Area for Each Bolt, Based on Max Load	134.6956	mm ²
Area Available in a Single Bolt (Corr)	338.0000	mm ²
Area Available in all the Bolts (Corr)	5407.9995	mm ²
Bolt Stress Based on Simplified Analysis	45.4	MPa
Allowable Bolt Stress	114.0 [Fact]	114.00 MPa

Shear Stress in a Single Bolt [tb]:
 $= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 51758.6 / (338.0 \cdot 16)$
 $= 9.6 \text{ MPa}$. Must be less than 107.9 MPa.

Concrete Contact Area of Base Ring	CCA 0.1043E+07	mm ²
Concrete Contact Section Modulus of Base Ring	0.5350E+09	mm ³

Concrete Load (Simplified method), Wind in Operating Condition [Sc]:
 $= ((ppl/100 \cdot (Abt \cdot Sa) + W)/Cca) + M/CZ$ per Jawad & Farr Eq. 12.1
 $= (1.0(5408.0 \cdot 114 + 206184)/1042908) + 0.23953E+09/53495E+09$
 $= 1.24 \text{ MPa}$

Concrete Allowable Stress	8.75 MPa
---------------------------	----------

Determine Maximum Bending Width of Basing Section [Rw1]:
 $= (Dou - SkirtOD)/2$
 $= (2336.0 - 2132.0)/2$
 $= 102.0 \text{ mm}$

Determine Maximum Bending Width of Basing Section [Rw2]:
 $= (SkirtID - Di + 2 \cdot Sca)/2$
 $= (2112.0 - 2032.0 + 2 \cdot 0.8)/2$
 $= 40.8 \text{ mm}$

Calculation of required Basing Thickness, (Simplified) [Tb]:
 $= \max(Rw1, Rw2) \cdot \sqrt{(3 \cdot Sc / S) + CA}$ per Jawad & Farr Eq. 12.12
 $= \max(102.0, 40.8001) \cdot \sqrt{(3 \cdot 1.236/174.663) + 1.5}$
 $= 16.3645 \text{ mm}$

Basing Stress at given Thickness [Sb]:
 $= 3 \cdot Sc(\max[Rw1, Rw2]/(Tb - Ca))^2$
 $= 3 \cdot 1.236(\max[102.0, 40.8]/(36.0 - 1.5))^2$
 $= 32.424$, must be less than 174.663 MPa

Required Thickness of Top Plate in Tension:
 (Calculated as a fixed beam per Megyesy)

$Ft = (Sa \cdot Abss)$,	Bolt Allowable Stress * Area
$Rm = (Ft \cdot 2 \cdot Cg)/8$,	Bending Moment
Sb	Allowable Bending Stress
$Wt = (Topwth - Bnd)$,	Width of Section

Required Thickness of Top Plate [T]:
 $= \sqrt{(6 \cdot Rm / (Sb \cdot Wt)) + CA}$
 $= \sqrt{(6 \cdot 433616 / (175 \cdot 95.0)) + 1.5}$
 $= 14.020 \text{ mm}$

Required Thickness of Continuous Top Ring per Moss:

$a = (Dc - SkirtOD)/2$	Skirt Distance to Bolt Circle
$P = Sa \cdot Abss$	Bolt Allowable Stress * Area
$l = Avgwdt$	Average Gusset Width



$g1 = \text{Gamma } 1$ Constant Term $f(b/l)$
 $g2 = \text{Gamma } 2$ Constant Term $f(b/l)$
 $g = \text{Flat distance } / 2$ Nut 1/2 Dimension (from Tema)
 $Fb =$ Allowable Bending Stress

$M_o = P/(4\pi)[1.3(\ln((2l\sin(\pi a/l)/(\pi g))) + 1] - [(0.7 - g2)P/(4\pi)]$ Moment Term

Required Thickness of Top Ring [T]:

$= \sqrt{(6 \cdot \text{Abs}(M_o) / F_b)} + CA$ Required Thickness

$= \sqrt{(6 \cdot 146951/175)} + 1.5$

$= 15.595 \text{ mm}$

Required Thickness of Gusset in Compression, per AISC E2-1:

1. Allowed Compression at Given Thickness:

Factor Kl/r Per E2-1	64.0848	
Factor C_c Per E2-1	123.2880	
Allowable Buckling Str. per E2-1	122.88	MPa
Actual Buckling Str. at Given Thickness	18.89	MPa

Required Gusset/Support Plate thickness, + CA 5.7487 mm

2. Allowed Compression at Calculated Thickness:

Factor Kl/r Per E2-1	150.8334	
Factor C_c Per E2-1	123.2880	
Allowable Buckling Str. per E2-2	45.66	MPa
Act. Buckling Str. at Calculated Thickness	44.46	MPa

Summary of Basing Thickness Calculations:

Required Basing Thickness (simplified)	16.3645	mm
Actual Basing Thickness	36.0000	mm

Required Top Ring/Plate Thickness as a Fixed Beam 14.0198 mm

Required Thickness of Continuous Top Ring (Moss) 15.5953 mm

Actual Top Plate/Ring Thickness 25.0000 mm

Required Gusset/Support Plate thickness, + CA 5.7487 mm

Actual Gusset/Support Plate Thickness 10.0000 mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$

$= 12.035 \cdot (10.0 - 1.6)$

$= 101.09 \text{ N/mm}$

Results for Computed Minimum Basing Weld Size [BWeld]:

$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$

$= 101.09 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707]$

$= 0.736 \text{ mm}$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$= \text{Bolt Load} / (Cmwh + 2 \cdot (Hg + Tta))$

$= 38528.0 / (110.0 + 2 \cdot (185.0 + 25.0))$

$= 72.694 \text{ N/mm}$

Horizontal Plate Load [Wh]:

$= \text{Bolt Load} \cdot e / (Cmwh \cdot (Hg + Tta) + 0.6667 \cdot (Hg + Tta)^2)$

$= 38528.0 \cdot 45.0 / (110.0 \cdot (210.0) + 0.6667 \cdot (210.0)^2)$

$= 33.024 \text{ N/mm}$

Resultant Weld Load [Wr]:

$= \sqrt{(Wv^2 + Wh^2)}$

$= \sqrt{(72.69^2 + 33.02^2)}$

$= 79.844 \text{ N/mm}$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$

$= 79.84 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707]$



= 0.581 mm

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$\begin{aligned} &= \text{Bolt Load} / (2 \cdot \text{TopWth}) \\ &= 38528.0 / (2 \cdot 102.0) \\ &= 188.863 \text{ N/mm} \end{aligned}$$

Weld Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth}) \\ &= 38528.0 \cdot 45.0 / (2 \cdot 210.0 \cdot 102.0) \\ &= 40.471 \text{ N/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{Wv^2 + Wh^2} \\ &= \sqrt{188.86^2 + 40.47^2} \\ &= 193.15 \text{ N/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 193.15 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707] \\ &= 1.405 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	4.7625	mm
Required Gusset to Skirt Double Fillet Weld Size	4.7625	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	1.4055	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N32"

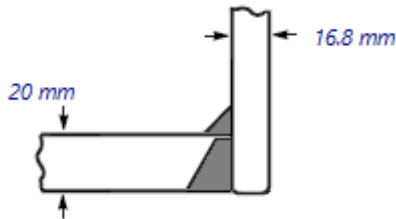
From: 20

Pressure for Reinforcement Calculations	P	2.3072	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	2100.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	20.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		15.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1020.0	mm
Centerline Distance to Tangent		700.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 2.31 \cdot 2103.0 \cdot 0.8988 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 2.31)$
 $= 15.8271 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)]) - 1)$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([2.31 / (138.0 \cdot 1.0)] - 1))$
 $= 0.4535 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3890 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	121.3280 mm
Parallel to Vessel Wall	Rn+tn+t	60.6640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N32".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9535 mm
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 19.0762 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 19.0762 mm
Wall Thickness, shell/head, external pressure	trb2	= 2.2707 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm
Wall Thickness per table UG-45	tb3	= 6.3000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(19.0762, 3.0)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9535, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
 (see 319.4.4 and 302.3.5):**

Sustained : 14.1, Allowable : 138.0 MPa Passed



Expansion : 0.0, Allowable : 330.9 MPa Passed
 Occasional : 1.6, Allowable : 183.5 MPa Passed
 Shear : 11.9, Allowable : 96.6 MPa Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 16.764, tr = 0.454, c = 1.5 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.03, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -13 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N32"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2640 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4664 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.8414 mm

Input Echo, WRC107/537 Item 1, Description: N32" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	3780.000	mm
Vessel Thickness	Tv	20.000	mm
Design Temperature	T1	80.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	50.800	mm
Nozzle Thickness	Tn	16.764	mm
Nozzle Material		SA-350 LF2	
Nozzle UNS Number		K03011	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Design Internal Pressure	Dp	2.307	MPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-1618.0	N
Longitudinal Shear	(SUS)	(V1) V1	2500.5	N
Circumferential Shear	(SUS)	(Vc) V2	2500.5	N

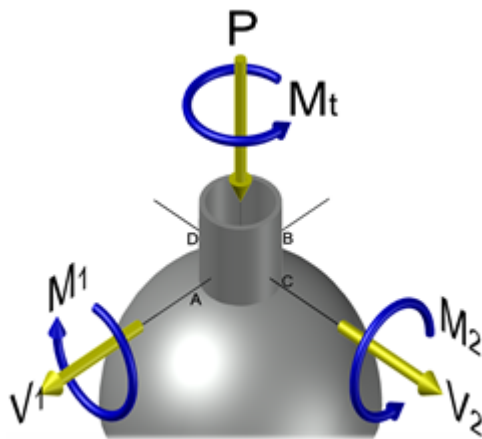


Circumferential Moment (SUS)	(Mc)	M1	343341.9	N-mm
Longitudinal Moment (SUS)	(Ml)	M2	539537.5	N-mm
Torsional Moment (SUS)		Mt	637636.0	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(1900.75 (20.0 - 1.5))}$$

$$= 703.145 \text{ mm}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load		P	-1618.0	N
Circumferential Shear	(VC)	V2	2500.5	N
Longitudinal Shear	(VL)	V1	2500.5	N
Circumferential Moment	(MC)	M1	343341.9	N-mm
Longitudinal Moment	(ML)	M2	539537.5	N-mm
Torsional Moment		MT	637636.0	N-mm

Dimensionless Param: U = 0.22 TAU = 5.00 (2.26) RHO = 1.21

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.12276	(A,B,C,D)
$M(x) / P$	SP 2	0.15408	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / MC$	SM 2	0.32281	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / MC$	SM 2	0.58892	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / ML$	SM 2	0.32281	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / ML$	SM 2	0.58892	(A,B,C,D)
$N(y) * T / P$	SP 2	0.19527	(A,B,C,D)
$M(y) / P$	SP 2	0.07342	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / MC$	SM 2	0.02675	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / MC$	SM 2	0.34960	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / ML$	SM 2	0.02675	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / ML$	SM 2	0.34960	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Rad. Bend. P		4.4	-4.4	4.4	-4.4	4.4	-4.4	4.4	-4.4
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.7	-1.7	1.7	1.7
Rad. Bend. MC		0.0	0.0	0.0	0.0	-18.9	18.9	18.9	-18.9
Rad. Memb. ML		-2.7	-2.7	2.7	2.7	0.0	0.0	0.0	0.0
Rad. Bend. ML		-29.7	29.7	29.7	-29.7	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-27.5	23.2	37.4	-30.8	-15.7	13.4	25.6	-21.0
Tang. Memb. P		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tang. Bend. P		2.1	-2.1	2.1	-2.1	2.1	-2.1	2.1	-2.1
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC		0.0	0.0	0.0	0.0	-11.2	11.2	11.2	-11.2
Tang. Memb. ML		-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Tang. Bend. ML		-17.6	17.6	17.6	-17.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-14.8	16.2	20.9	-18.6	-8.4	9.9	14.4	-12.2
Shear VC		1.0	1.0	-1.0	-1.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT		3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Tot. Shear		4.1	4.1	2.1	2.1	2.1	2.1	4.1	4.1
Str. Int.		28.7	25.1	37.6	31.1	16.2	14.3	26.9	22.6

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		118.0	118.0	118.0	118.0	118.0	118.0	118.0	118.0
Rad. Pl (SUS)		-2.1	-2.1	3.3	3.3	-1.1	-1.1	2.3	2.3
Rad. Q (SUS)		-25.3	25.3	34.1	-34.1	-14.5	14.5	23.3	-23.3
Long. Pm (SUS)		118.0	118.0	118.0	118.0	118.0	118.0	118.0	118.0
Long. Pl (SUS)		0.7	0.7	1.1	1.1	0.8	0.8	1.1	1.1
Long. Q (SUS)		-15.5	15.5	19.7	-19.7	-9.1	9.1	13.3	-13.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.0	1.0	-1.0	-1.0	-1.0	-1.0	1.0	1.0
Shear Q (SUS)		3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Pm (SUS)		118.0	118.0	118.0	118.0	118.0	118.0	118.0	118.0
Pm+Pl (SUS)		119.0	119.0	121.7	121.7	119.2	119.2	120.8	120.8
Pm+Pl+Q (Total)		104.3	143.1	155.6	99.7	110.1	132.3	144.9	107.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	117.95	138.00	Passed
Pm+Pl (SUS)	121.66	207.00	Passed
Pm+Pl+Q (TOTAL)	155.57	413.99	Passed



*Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.*

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Input, Nozzle Desc: MH24"

From: 30

Pressure for Reinforcement Calculations	P	2.3001	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		750.00	mm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

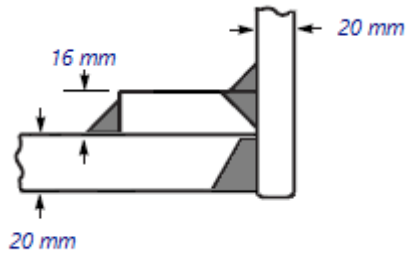
Outside Projection	ho	167.3760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1219.2000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		284.8000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: MH24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 17.7034 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 306.3 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 5.1569 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1714 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.2000 mm
Parallel to Vessel Wall, opening length	d	612.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.2500 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

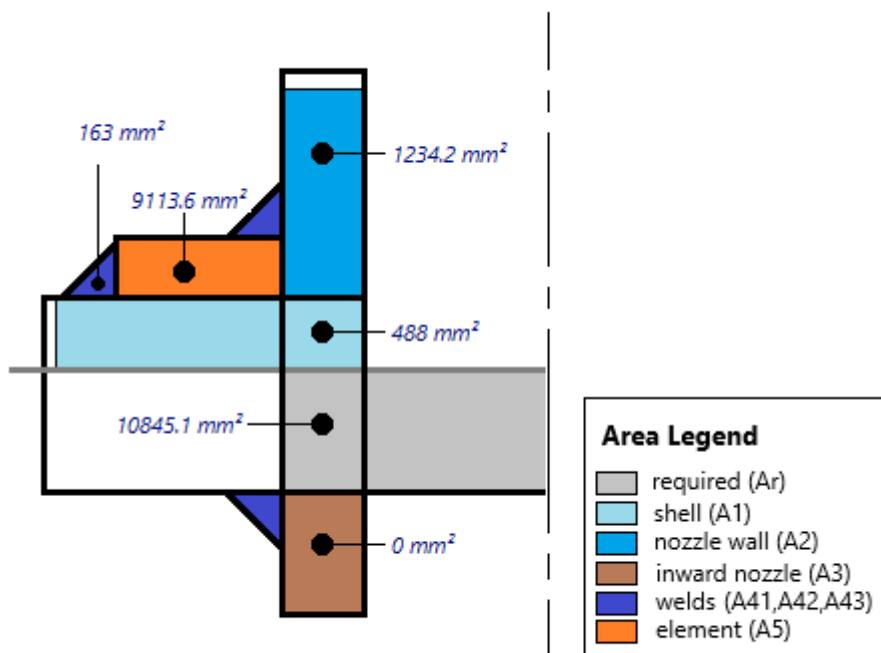


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10845.091	2849.004	NA
Area in Shell	A1	488.009	5635.091	NA
Area in Nozzle Wall	A2	1234.240	1602.897	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	163.000	163.000	NA
Area in Element	A5	9113.601	9113.601	NA
TOTAL AREA AVAILABLE	Atot	10998.849	16514.590	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	1209.5901	16.0000 mm
Based on given Pad Diameter:	1219.2000	15.7301 mm
Based on Shell or Nozzle Thickness:	1097.5920	20.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (612.6 \cdot 17.7034 \cdot 1.0 + 2 \cdot 18.5 \cdot 17.7034 \cdot 1.0 \cdot (1 - 1.0))$
 $= 10845.091 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 612.6(1.0 \cdot 18.5 - 1.0 \cdot 17.703) - 2 \cdot 18.5$
 $(1.0 \cdot 18.5 - 1.0 \cdot 17.7034) \cdot (1 - 1.0)$
 $= 488.009 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 46.25) \cdot (18.5 - 5.16) \cdot 1.0$



= 1234.240 mm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (100.0) • 1.0 + (0.0) • 1.0 + 63.0003² • 1.0
= 163.000 mm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (1219.2 - 649.6) • 16.0 • 1.0
= 9113.601 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures ta = 6.6569 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= ta
= 6.6569 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16.0, c = 1.5 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.957, Temp. Reduction = 2 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-16 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 16.0, c = 1.5 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.957, Temp. Reduction = 2 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-16 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 20.0, tr = 17.703, c = 1.5 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.957, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-10 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-29 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-85 °C



Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 2.30/5.10 = 0.451

Weld Size Calculations, Description: MH24"

Intermediate Calc. for nozzle/shell Welds Tmin 16.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 16.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	8.0000 = 0.5*TminPad	8.4840 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (10845.0908 - 488.0088 + 2 \cdot 18.5 \cdot 1.0 \cdot (1.0 \cdot 18.5 - 17.7034)) \cdot 138)$
 $= 1433197.12 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (1234.2397 + 9113.6006 + 163.0003 - 0.0 \cdot 1.0) \cdot 138$
 $= 1450346.38 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (1234.2397 + 0.0 + 100.0 + (684.4999)) \cdot 138$
 $= 278557.34 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (1234.2397 + 0.0 + 163.0003 + 9113.6006 + (684.4999)) \cdot 138$
 $= 1544797.75 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 649.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 689928. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1219.2 \cdot 12.0 \cdot 0.49 \cdot 138$
 $= 1553840. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 315.55) \cdot (20.0 - 1.5) \cdot 0.7 \cdot 138$
 $= 1771454. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 649.6 \cdot 16.0 \cdot 0.74 \cdot 138$
 $= 1667093. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 649.6 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 1667093. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:



Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1553841 + 1771454) = 3325295 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (689929 + 1667093 + 1667093 + 0) = 4024114 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (1553841 + 1667093 + 0) = 3220933 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 3325294 N , must exceed W = 1433197 N or W1 = 1450346 N
Path 2-2 = 4024113 N , must exceed W = 1433197 N or W2 = 278557 N
Path 3-3 = 3220933 N , must exceed W = 1433197 N or W3 = 1544797 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 51.4986 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 238.8747 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.177 %

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Input, Nozzle Desc: K1B3"

From: 30

Pressure for Reinforcement Calculations	P	2.3031	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		150.00	mm
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User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

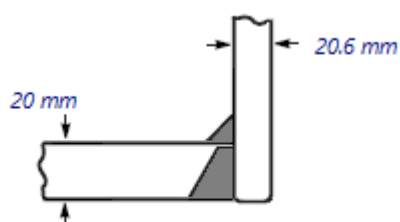
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	307.0760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K1B3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 17.7268 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 0.6676 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5574 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm
Parallel to Vessel Wall, opening length	d	79.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

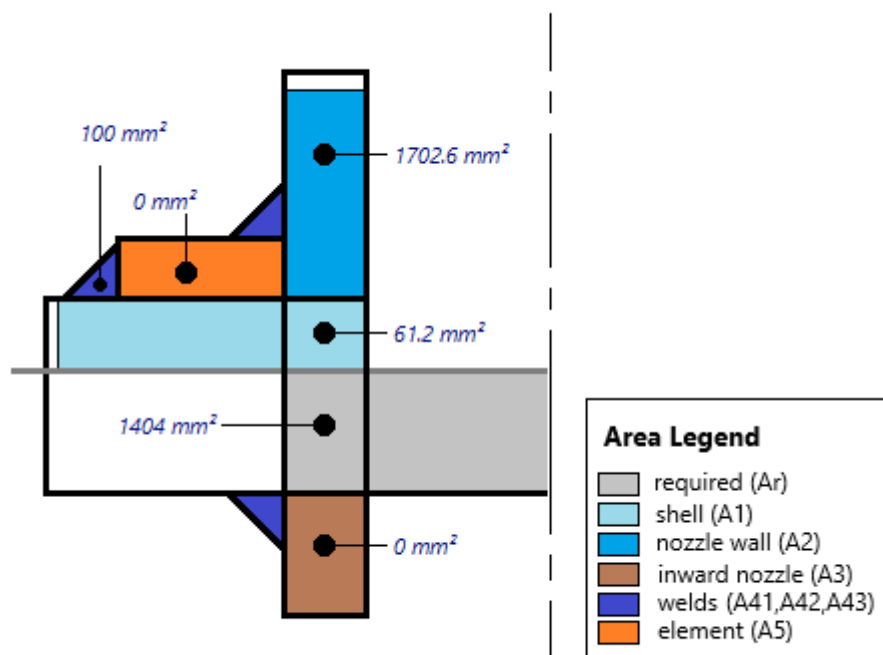


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	1403.959	368.334	NA
Area in Shell	A1	61.241	728.533	NA
Area in Nozzle Wall	A2	1702.593	1712.782	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.834	2541.315	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 17.7268 \cdot 1.0 + 2 \cdot 19.074 \cdot 17.7268 \cdot 1.0 \cdot (1 - 1.0))$
 $= 1403.959 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 18.5 - 1.0 \cdot 17.727) - 2 \cdot 19.074$
 $(1.0 \cdot 18.5 - 1.0 \cdot 17.7268) \cdot (1 - 1.0)$
 $= 61.241 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 46.25)(19.07 - 0.67)1.0$
 $= 1702.593 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1676 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 19.2268 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 19.2268 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.2724 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(19.2268, 3.0)]$
 $= 7.2300 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1676, 7.23)$
 $= 7.2300 \text{ mm}$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 20.0$, $tr = 17.727$, $c = 1.5 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E*/(tg - c) = 0.958$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -10 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K1B3"

Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (1403.9586 - 61.2412 + 2 \cdot 19.074 \cdot 1.0 \cdot (1.0 \cdot 18.5 - 17.7268)) \cdot 138)$
 $= 189346.16 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (1702.5931 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 248732.19 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (1702.5931 + 0.0 + 100.0 + (705.7379)) \cdot 138$
 $= 346114.00 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (1702.5931 + 0.0 + 100.0 + 0.0 + (705.7379)) \cdot 138$
 $= 346114.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 117.348 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 124631. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 49.137) \cdot (20.574 - 1.5) \cdot 0.7 \cdot 138$
 $= 284407. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 117.348 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 301154. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (124631 + 284407) = 409038 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (124631 + 0 + 301155 + 0) = 425786 \text{ N}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (124631 + 301155 + 0) = 425786 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 409038 N , must exceed W = 189346 N or W1 = 248732 N



Path 2-2 = 425785 N , must exceed W = 189346 N or W2 = 346114 N
Path 3-3 = 425785 N , must exceed W = 189346 N or W3 = 346114 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.6406 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 328.7166 mm

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Input, Nozzle Desc: K2B3"

From: 30

Pressure for Reinforcement Calculations	P	2.3031	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		150.00	mm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

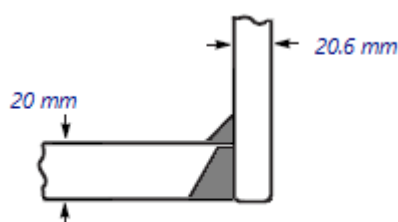
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	307.0760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2B3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 17.7268 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 0.6676 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5574 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm
Parallel to Vessel Wall, opening length	d	79.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

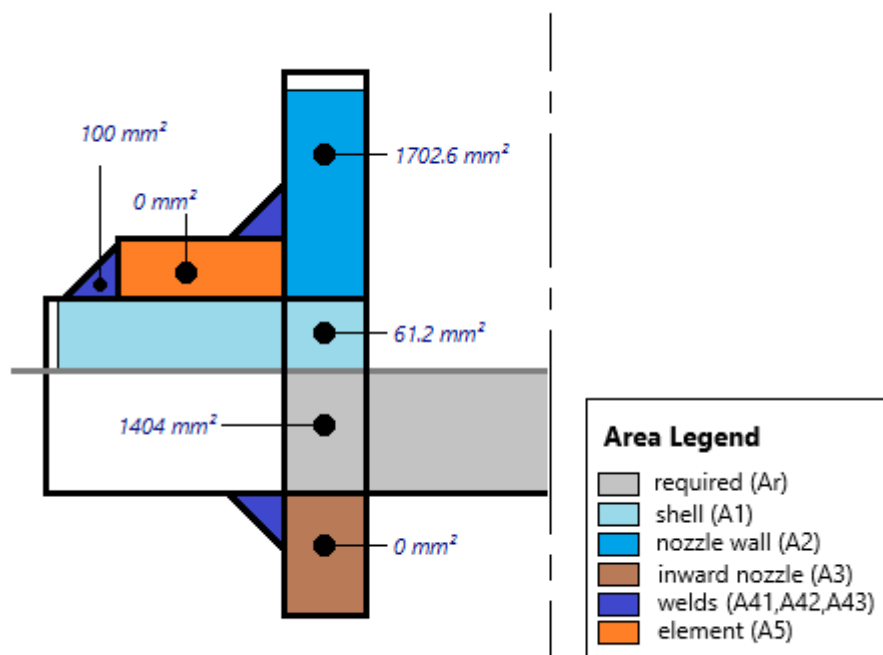


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	1403.959	368.334	NA
Area in Shell	A1	61.241	728.533	NA
Area in Nozzle Wall	A2	1702.593	1712.782	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.834	2541.315	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$
 $= (79.2 \cdot 17.7268 \cdot 1.0 + 2 \cdot 19.074 \cdot 17.7268 \cdot 1.0 \cdot (1 - 1.0))$
 $= 1403.959 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 18.5 - 1.0 \cdot 17.727) - 2 \cdot 19.074$
 $(1.0 \cdot 18.5 - 1.0 \cdot 17.7268) \cdot (1 - 1.0)$
 $= 61.241 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 46.25)(19.07 - 0.67)1.0$
 $= 1702.593 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1676 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 19.2268 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 19.2268 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.2724 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(19.2268, 3.0)]$
 $= 7.2300 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1676, 7.23)$
 $= 7.2300 \text{ mm}$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 20.0$, $tr = 17.727$, $c = 1.5 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.958$, Temp. Reduction = $2 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B $-7 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-10 \text{ }^\circ\text{C}$
 Min Metal Temp. w/o impact per UG-20(f) $-29 \text{ }^\circ\text{C}$



Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K2B3"

Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (1403.9586 - 61.2412 + 2 • 19.074 • 1.0 •
(1.0 • 18.5 - 17.7268))138)
= 189346.16 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1702.5931 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 248732.19 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1702.5931 + 0.0 + 100.0 + (705.7379)) • 138
= 346114.00 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1702.5931 + 0.0 + 100.0 + 0.0 + (705.7379)) • 138
= 346114.00 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 124631. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 284407. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (16.0 - 0.0) • 0.74 • 138
= 301154. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (124631 + 284407) = 409038 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (124631 + 0 + 301155 + 0) = 425786 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (124631 + 301155 + 0) = 425786 N

Summary of Failure Path Calculations:

Path 1-1 = 409038 N , must exceed W = 189346 N or W1 = 248732 N



Path 2-2 = 425785 N , must exceed W = 189346 N or W2 = 346114 N
Path 3-3 = 425785 N , must exceed W = 189346 N or W3 = 346114 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.6406 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 328.7166 mm

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Input, Nozzle Desc: K2D3"

From: 30

Pressure for Reinforcement Calculations	P	2.3031	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		150.00	mm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

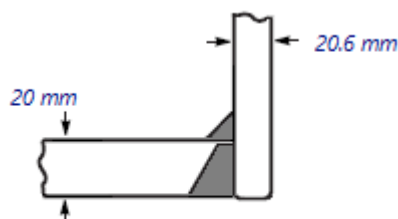
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	307.0760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2D3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 17.7268 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 0.6676 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5574 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm
Parallel to Vessel Wall, opening length	d	79.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

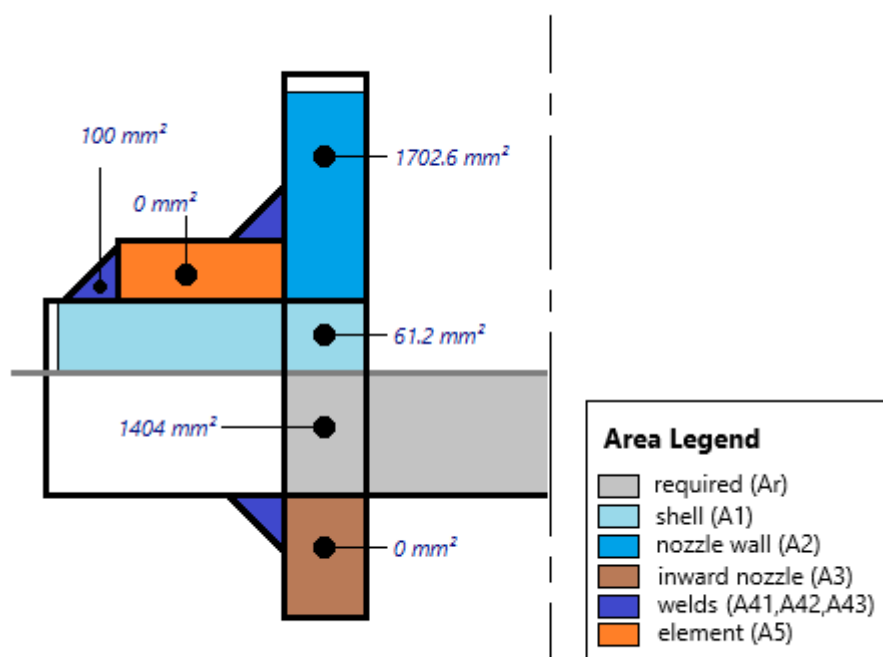


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	1403.959	368.334	NA
Area in Shell	A1	61.241	728.533	NA
Area in Nozzle Wall	A2	1702.593	1712.782	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.834	2541.315	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 17.7268 \cdot 1.0 + 2 \cdot 19.074 \cdot 17.7268 \cdot 1.0 \cdot (1 - 1.0))$
 $= 1403.959 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 18.5 - 1.0 \cdot 17.727) - 2 \cdot 19.074$
 $(1.0 \cdot 18.5 - 1.0 \cdot 17.7268) \cdot (1 - 1.0)$
 $= 61.241 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 46.25)(19.07 - 0.67)1.0$
 $= 1702.593 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1676 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 19.2268 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 19.2268 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.2724 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(19.2268, 3.0)]$
 $= 7.2300 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1676, 7.23)$
 $= 7.2300 \text{ mm}$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 20.0$, $tr = 17.727$, $c = 1.5 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E*/(tg - c) = 0.958$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -10 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K2D3"

Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (1403.9586 - 61.2412 + 2 • 19.074 • 1.0 •
(1.0 • 18.5 - 17.7268))138)
= 189346.16 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1702.5931 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 248732.19 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1702.5931 + 0.0 + 100.0 + (705.7379)) • 138
= 346114.00 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1702.5931 + 0.0 + 100.0 + 0.0 + (705.7379)) • 138
= 346114.00 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 124631. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 284407. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (16.0 - 0.0) • 0.74 • 138
= 301154. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (124631 + 284407) = 409038 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (124631 + 0 + 301155 + 0) = 425786 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (124631 + 301155 + 0) = 425786 N

Summary of Failure Path Calculations:

Path 1-1 = 409038 N , must exceed W = 189346 N or W1 = 248732 N

PV Elite 28 U2 Licensee: SL User
FileName : 401-V-32002_ - 7%
Nozzle Calcs.: K2D_3"

Nozl: 37 5:15pm Aug 11,2026

Page 96 of 168



HEXAGON

Path 2-2 = 425785 N , must exceed W = 189346 N or W2 = 346114 N
Path 3-3 = 425785 N , must exceed W = 189346 N or W3 = 346114 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.6406 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 328.7166 mm

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Input, Nozzle Desc: K2F3"

From: 30

Pressure for Reinforcement Calculations	P	2.3031	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		150.00	mm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		125.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

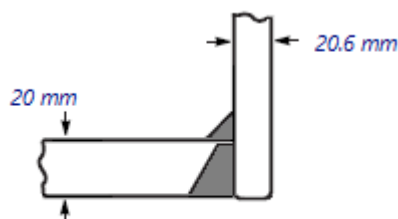
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	307.0760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2F3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 17.7268 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 0.6676 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5574 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm
Parallel to Vessel Wall, opening length	d	79.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

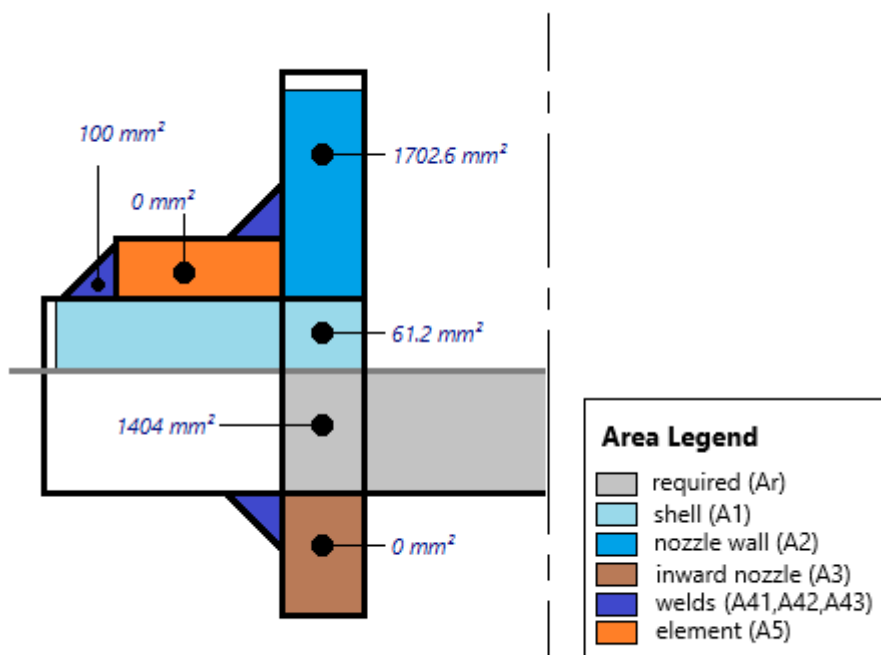


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	1403.959	368.334	NA
Area in Shell	A1	61.241	728.533	NA
Area in Nozzle Wall	A2	1702.593	1712.782	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.834	2541.315	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 17.7268 \cdot 1.0 + 2 \cdot 19.074 \cdot 17.7268 \cdot 1.0 \cdot (1 - 1.0))$
 $= 1403.959 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 18.5 - 1.0 \cdot 17.727) - 2 \cdot 19.074$
 $(1.0 \cdot 18.5 - 1.0 \cdot 17.7268) \cdot (1 - 1.0)$
 $= 61.241 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 46.25)(19.07 - 0.67)1.0$
 $= 1702.593 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1676 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 19.2268 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 19.2268 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.2724 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(19.2268, 3.0)]$
 $= 7.2300 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1676, 7.23)$
 $= 7.2300 \text{ mm}$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 20.0$, $tr = 17.727$, $c = 1.5 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E*/(tg - c) = 0.958$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -10 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K2F3"

Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (1403.9586 - 61.2412 + 2 • 19.074 • 1.0 •
(1.0 • 18.5 - 17.7268))138)
= 189346.16 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1702.5931 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 248732.19 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1702.5931 + 0.0 + 100.0 + (705.7379)) • 138
= 346114.00 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1702.5931 + 0.0 + 100.0 + 0.0 + (705.7379)) • 138
= 346114.00 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 124631. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 284407. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (16.0 - 0.0) • 0.74 • 138
= 301154. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (124631 + 284407) = 409038 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (124631 + 0 + 301155 + 0) = 425786 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (124631 + 301155 + 0) = 425786 N

Summary of Failure Path Calculations:

Path 1-1 = 409038 N , must exceed W = 189346 N or W1 = 248732 N



Path 2-2 = 425785 N , must exceed W = 189346 N or W2 = 346114 N
Path 3-3 = 425785 N , must exceed W = 189346 N or W3 = 346114 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.6406 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 328.7166 mm

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Input, Nozzle Desc: K1A3'

From: 30

Pressure for Reinforcement Calculations	P	2.3001	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		750.00	mm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

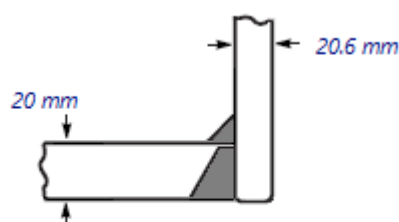
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	307.0760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K1A3'

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.810 in.

Nozzle input data check completed without errors.

$$\begin{aligned} & \text{Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]} \\ & = P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ & = 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3) \\ & = 17.7034 \text{ mm} \end{aligned}$$
$$\begin{aligned} & \text{Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]} \\ & = P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ & = 2.3 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 2.3) \\ & = 0.6667 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.5574 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm
Parallel to Vessel Wall, opening length	d	79.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Weld Strength Reduction Factor [fr1]:
 = min(1, Sn/Sv)
 = min(1, 138.0/138.0)
 = 1.000

Weld Strength Reduction Factor [fr2]:
 = min(1, Sn/Sv)
 = min(1, 138.0/138.0)
 = 1.000

Weld Strength Reduction Factor [fr3]:
 = min(fr2, fr4)
 = min(1.0, 1.0)
 = 1.000

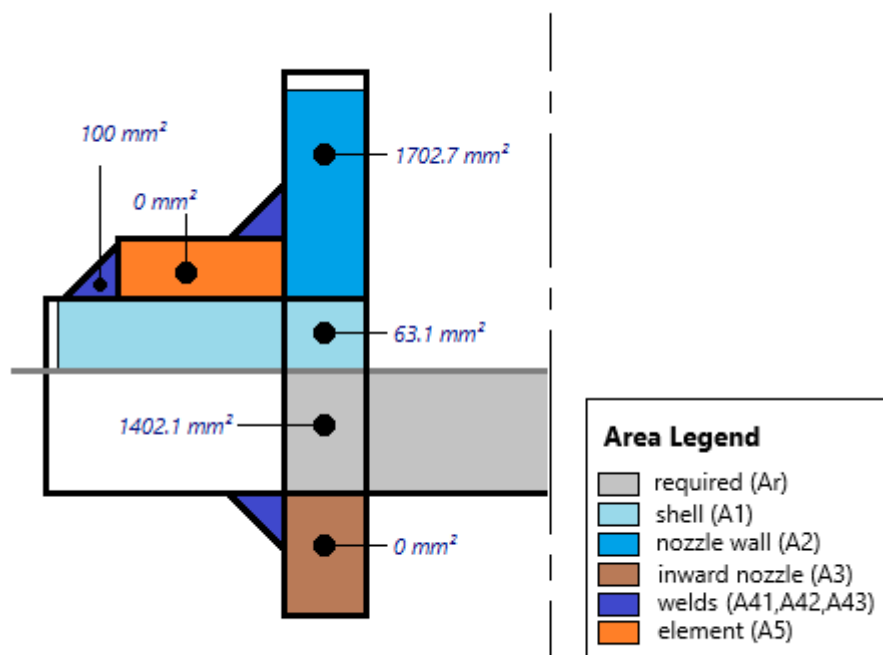


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	1402.108	368.334	NA
Area in Shell	A1	63.092	728.533	NA
Area in Nozzle Wall	A2	1702.674	1712.782	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1865.767	2541.315	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 17.7034 \cdot 1.0 + 2 \cdot 19.074 \cdot 17.7034 \cdot 1.0 \cdot (1 - 1.0))$
 $= 1402.108 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 18.5 - 1.0 \cdot 17.703) - 2 \cdot 19.074$
 $(1.0 \cdot 18.5 - 1.0 \cdot 17.7034) \cdot (1 - 1.0)$
 $= 63.092 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 46.25)(19.07 - 0.67)1.0$
 $= 1702.674 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1667 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 19.2034 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 19.2034 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.2724 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(19.2034, 3.0)]$
 $= 7.2300 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1667, 7.23)$
 $= 7.2300 \text{ mm}$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 20.0$, $tr = 17.703$, $c = 1.5 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.957$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -10 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K1A3'

Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (1402.1078 - 63.0922 + 2 • 19.074 • 1.0 •
(1.0 • 18.5 - 17.7034))•138)
= 188958.39 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1702.6744 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 248743.42 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1702.6744 + 0.0 + 100.0 + (705.7379)) • 138
= 346125.22 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1702.6744 + 0.0 + 100.0 + 0.0 + (705.7379)) • 138
= 346125.22 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 124631. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 284407. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (16.0 - 0.0) • 0.74 • 138
= 301154. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (124631 + 284407) = 409038 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (124631 + 0 + 301155 + 0) = 425786 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (124631 + 301155 + 0) = 425786 N

Summary of Failure Path Calculations:

Path 1-1 = 409038 N , must exceed W = 188958 N or W1 = 248743 N



Path 2-2 = 425785 N , must exceed W = 188958 N or W2 = 346125 N
Path 3-3 = 425785 N , must exceed W = 188958 N or W3 = 346125 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.6406 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 328.7166 mm

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Input, Nozzle Desc: K2A3"

From: 30

Pressure for Reinforcement Calculations	P	2.3001	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		750.00	mm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

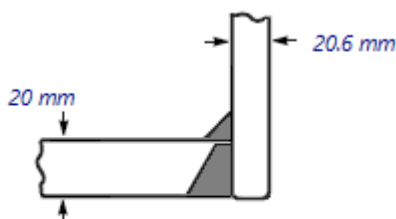
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	307.0760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2A3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 17.7034 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 0.6667 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5574 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm
Parallel to Vessel Wall, opening length	d	79.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

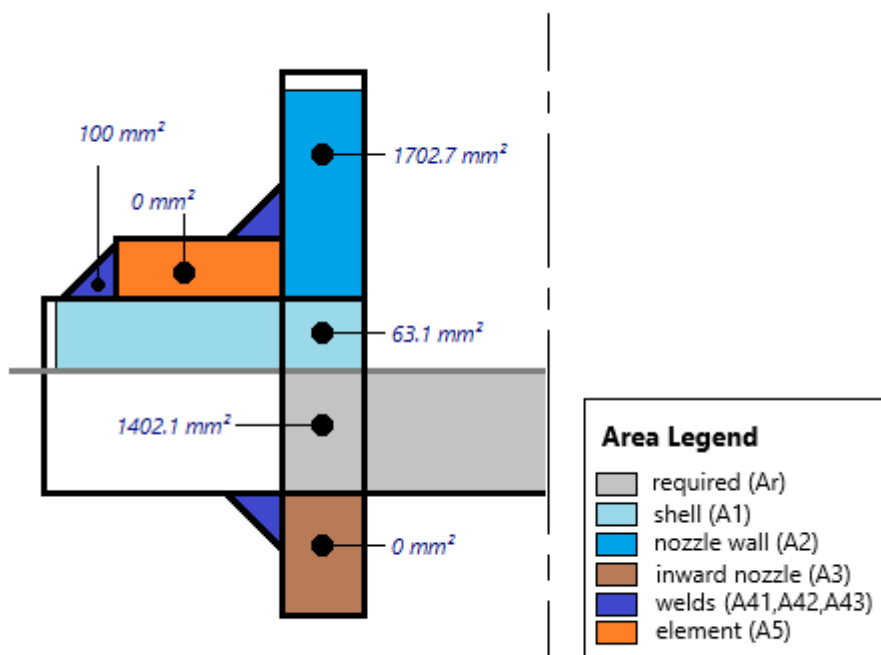


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	1402.108	368.334	NA
Area in Shell	A1	63.092	728.533	NA
Area in Nozzle Wall	A2	1702.674	1712.782	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1865.767	2541.315	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 17.7034 \cdot 1.0 + 2 \cdot 19.074 \cdot 17.7034 \cdot 1.0 \cdot (1 - 1.0))$
 $= 1402.108 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 18.5 - 1.0 \cdot 17.703) - 2 \cdot 19.074$
 $(1.0 \cdot 18.5 - 1.0 \cdot 17.7034) \cdot (1 - 1.0)$
 $= 63.092 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 46.25)(19.07 - 0.67)1.0$
 $= 1702.674 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1667 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 19.2034 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 19.2034 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.2724 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(19.2034, 3.0)]$
 $= 7.2300 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1667, 7.23)$
 $= 7.2300 \text{ mm}$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 20.0$, $tr = 17.703$, $c = 1.5 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.957$, Temp. Reduction = $2 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B $-7 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-10 \text{ }^\circ\text{C}$
 Min Metal Temp. w/o impact per UG-20(f) $-29 \text{ }^\circ\text{C}$



Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K2A3"

Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (1402.1078 - 63.0922 + 2 \cdot 19.074 \cdot 1.0 \cdot \\
 &\quad (1.0 \cdot 18.5 - 17.7034)) \cdot 138) \\
 &= 188958.39 \text{ N}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (1702.6744 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138 \\
 &= 248743.42 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (1702.6744 + 0.0 + 100.0 + (705.7379)) \cdot 138 \\
 &= 346125.22 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (1702.6744 + 0.0 + 100.0 + 0.0 + (705.7379)) \cdot 138 \\
 &= 346125.22 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 117.348 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 124631. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 49.137) \cdot (20.574 - 1.5) \cdot 0.7 \cdot 138 \\
 &= 284407. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 117.348 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138 \\
 &= 301154. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (124631 + 284407) = 409038 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (124631 + 0 + 301155 + 0) = 425786 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (124631 + 301155 + 0) = 425786 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 409038 N , must exceed W = 188958 N or W1 = 248743 N



Path 2-2 = 425785 N , must exceed W = 188958 N or W2 = 346125 N
Path 3-3 = 425785 N , must exceed W = 188958 N or W3 = 346125 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.6406 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 328.7166 mm

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Input, Nozzle Desc: K2C3"

From: 30

Pressure for Reinforcement Calculations	P	2.3001	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		750.00	mm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

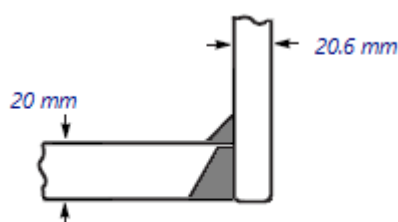
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	307.0760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2C3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 17.7034 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 0.6667 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5574 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm
Parallel to Vessel Wall, opening length	d	79.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

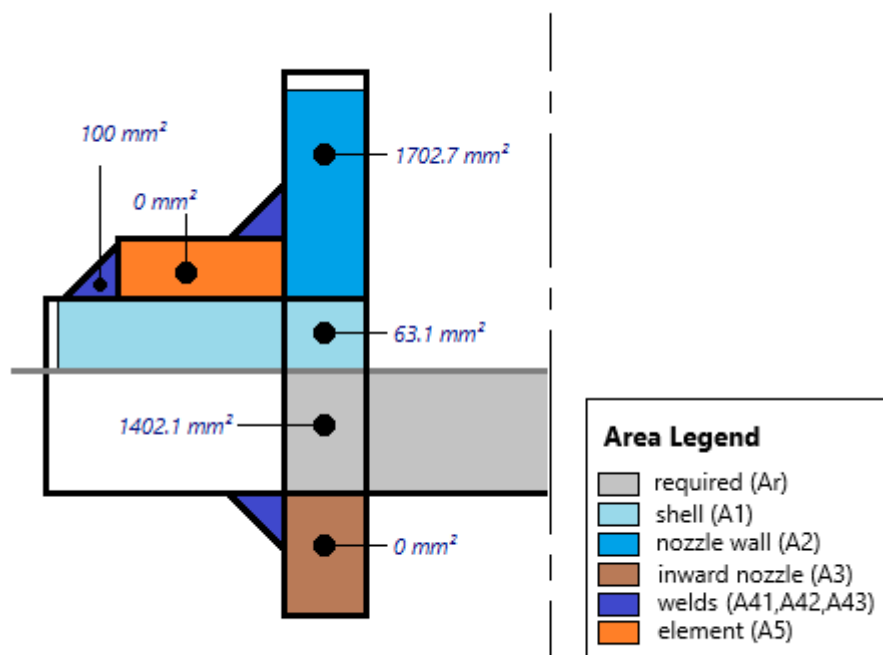


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
--------------------------	--------	----------	-------



Area Required	Ar	1402.108	368.334	NA
Area in Shell	A1	63.092	728.533	NA
Area in Nozzle Wall	A2	1702.674	1712.782	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1865.767	2541.315	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 17.7034 \cdot 1.0 + 2 \cdot 19.074 \cdot 17.7034 \cdot 1.0 \cdot (1 - 1.0))$
 $= 1402.108 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 18.5 - 1.0 \cdot 17.703) - 2 \cdot 19.074$
 $(1.0 \cdot 18.5 - 1.0 \cdot 17.7034) \cdot (1 - 1.0)$
 $= 63.092 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 46.25)(19.07 - 0.67)1.0$
 $= 1702.674 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1667 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 19.2034 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 19.2034 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.2724 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(19.2034, 3.0)]$
 $= 7.2300 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1667, 7.23)$
 $= 7.2300 \text{ mm}$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 20.0$, $tr = 17.703$, $c = 1.5 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.957$, Temp. Reduction = $2 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B $-7 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-10 \text{ }^\circ\text{C}$
 Min Metal Temp. w/o impact per UG-20(f) $-29 \text{ }^\circ\text{C}$



Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K2C3"

Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (1402.1078 - 63.0922 + 2 • 19.074 • 1.0 •
(1.0 • 18.5 - 17.7034))•138)
= 188958.39 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1702.6744 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 248743.42 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1702.6744 + 0.0 + 100.0 + (705.7379)) • 138
= 346125.22 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1702.6744 + 0.0 + 100.0 + 0.0 + (705.7379)) • 138
= 346125.22 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 124631. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 284407. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (16.0 - 0.0) • 0.74 • 138
= 301154. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (124631 + 284407) = 409038 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (124631 + 0 + 301155 + 0) = 425786 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (124631 + 301155 + 0) = 425786 N

Summary of Failure Path Calculations:

Path 1-1 = 409038 N , must exceed W = 188958 N or W1 = 248743 N



Path 2-2 = 425785 N , must exceed W = 188958 N or W2 = 346125 N
Path 3-3 = 425785 N , must exceed W = 188958 N or W3 = 346125 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.6406 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 328.7166 mm

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Input, Nozzle Desc: K2E3"

From: 30

Pressure for Reinforcement Calculations	P	2.3001	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		750.00	mm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		125.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

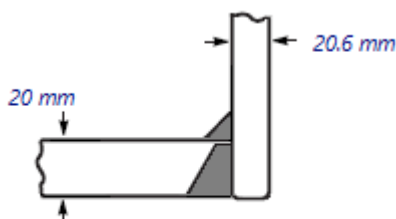
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	307.0760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2E3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 17.7034 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 0.6667 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5574 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm
Parallel to Vessel Wall, opening length	d	79.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

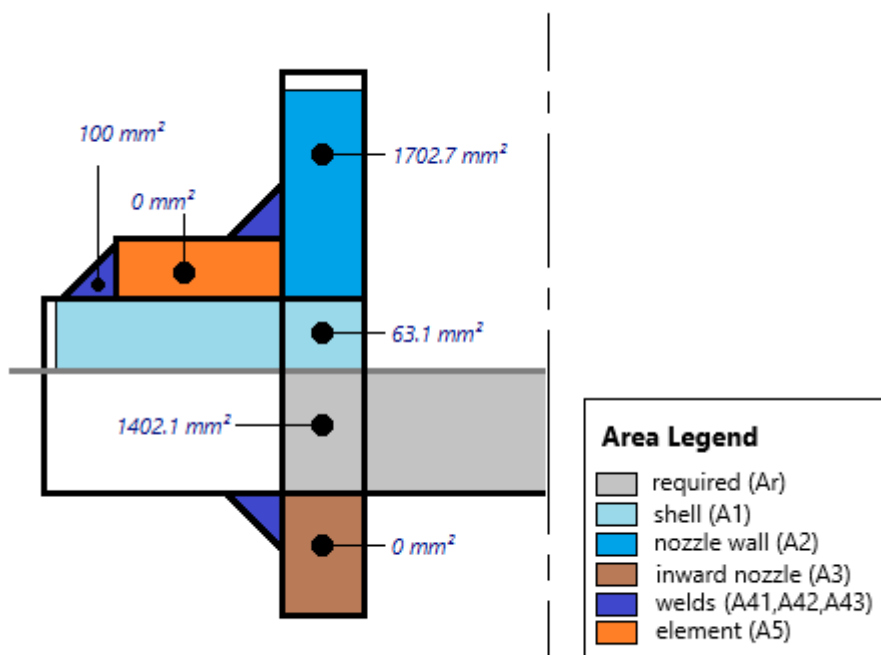


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	1402.108	368.334	NA
Area in Shell	A1	63.092	728.533	NA
Area in Nozzle Wall	A2	1702.674	1712.782	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1865.767	2541.315	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 17.7034 \cdot 1.0 + 2 \cdot 19.074 \cdot 17.7034 \cdot 1.0 \cdot (1 - 1.0))$
 $= 1402.108 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 18.5 - 1.0 \cdot 17.703) - 2 \cdot 19.074$
 $(1.0 \cdot 18.5 - 1.0 \cdot 17.7034) \cdot (1 - 1.0)$
 $= 63.092 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 46.25)(19.07 - 0.67)1.0$
 $= 1702.674 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1667 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 19.2034 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 19.2034 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.2724 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(19.2034, 3.0)]$
 $= 7.2300 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1667, 7.23)$
 $= 7.2300 \text{ mm}$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 20.0$, $tr = 17.703$, $c = 1.5 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.957$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -10 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K2E3"

Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (1402.1078 - 63.0922 + 2 • 19.074 • 1.0 •
(1.0 • 18.5 - 17.7034))•138)
= 188958.39 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1702.6744 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 248743.42 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1702.6744 + 0.0 + 100.0 + (705.7379)) • 138
= 346125.22 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1702.6744 + 0.0 + 100.0 + 0.0 + (705.7379)) • 138
= 346125.22 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 124631. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 284407. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (16.0 - 0.0) • 0.74 • 138
= 301154. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (124631 + 284407) = 409038 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (124631 + 0 + 301155 + 0) = 425786 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (124631 + 301155 + 0) = 425786 N

Summary of Failure Path Calculations:

Path 1-1 = 409038 N , must exceed W = 188958 N or W1 = 248743 N



Path 2-2 = 425785 N , must exceed W = 188958 N or W2 = 346125 N
Path 3-3 = 425785 N , must exceed W = 188958 N or W3 = 346125 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.6406 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 328.7166 mm

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Input, Nozzle Desc: K3A1.5"

From: 30

Pressure for Reinforcement Calculations	P	2.3001	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	400.0000	mm
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Distance from Bottom/Left Tangent		750.00	mm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		240.65	deg
Diameter		1.5000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm

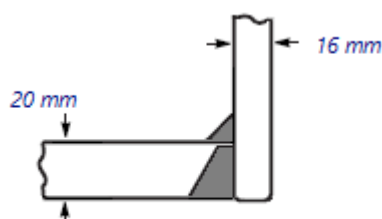
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	207.9620	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.



Reinforcement CALCULATION, Description: K3A1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500 in.
Actual Thickness Used in Calculation	0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
= 17.7034 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $20.55 \cdot (\exp([2.3 / (138.0 \cdot 1.0)] - 1))$
= 0.3454 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3519 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	110.3877 mm
Parallel to Vessel Wall	Rn+tn+t	55.1938 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3A1.5".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.002, tr = 0.345, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.024$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K3A1.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	14.5020 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K3A1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500 in.
Actual Thickness Used in Calculation	0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$$

$$= 17.7034 \text{ mm}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R (\exp([P / (S_n \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 20.55 (\exp([2.3 / (138.0 \cdot 1.0)] - 1))$$

$$= 0.3454 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3519 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.1040	mm
Parallel to Vessel Wall	Rn+tn+t	53.5520	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K3A1.5".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8519	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 19.2034	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 19.2034	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.2724	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.0200	mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.02, \max(19.2034, 3.0)]$$

$$= 6.0200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.8519, 6.02)$$

$$= 6.0200 \text{ mm}$$

Available Nozzle Neck Thickness = 16.0020 mm -> OK

Weld Size Calculations, Description: K3A1.5"

Intermediate Calc. for nozzle/shell Welds Tmin 14.5020 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 15.1938 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 244.7528 mm

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Input, Nozzle Desc: K3B1.5"

From: 30

Pressure for Reinforcement Calculations	P	2.3031	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	400.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		150.00	mm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		240.65	deg
Diameter		1.5000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm

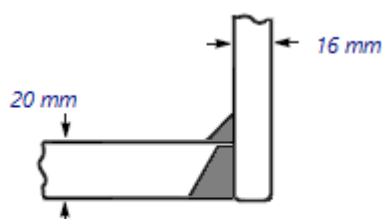
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	207.9620	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.



Reinforcement CALCULATION, Description: K3B1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500 in.
Actual Thickness Used in Calculation	0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
= 17.7268 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $20.55 \cdot (\exp([2.3 / (138.0 \cdot 1.0)] - 1))$
= 0.3458 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3519 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	110.3876 mm
Parallel to Vessel Wall	Rn+tn+t	55.1938 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3E1.5".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.002, tr = 0.346, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.024$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K3B1.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	14.5020 mm
---	------	------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K3B1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500 in.
Actual Thickness Used in Calculation	0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$\begin{aligned} &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3) \\ &= 17.7268 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{Reqd thk per App. 1 of Nozzle Wall, trn} \quad [\text{Int. Press}] \\ &= R \left(\exp([P / (S_n \cdot E)] - 1) \right) \text{ per Appendix 1-2 (a)(1)} \\ &= 20.55 \left(\exp([2.3 / (138.0 \cdot 1.0)] - 1) \right) \\ &= 0.3458 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3519 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.1040	mm
Parallel to Vessel Wall	Rn+tn+t	53.5520	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3B1.5".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8519	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 19.2268	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 19.2268	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.2724	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.0200	mm

$$\begin{aligned} &\text{Determine Nozzle Thickness candidate [tb]:} \\ &= \min[tb3, \max(tb1, tb2)] \\ &= \min[6.02, \max(19.2268, 3.0)] \\ &= 6.0200 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{Minimum Wall Thickness of Nozzle Necks [tUG-45]:} \\ &= \max(ta, tb) \\ &= \max(1.8519, 6.02) \\ &= 6.0200 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 16.0020 mm -> OK

Weld Size Calculations, Description: K3B1.5"

$$\text{Intermediate Calc. for nozzle/shell Welds} \quad T_{\min} \quad 14.5020 \text{ mm}$$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 15.1938 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 244.7528 mm

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Input, Nozzle Desc: N42"

From: 30

Pressure for Reinforcement Calculations	P	2.3028	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		200.00	mm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		145.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

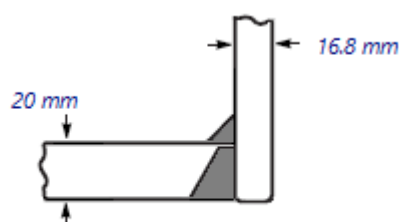
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	313.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
= 17.7244 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $26.9 \cdot (\exp([2.3 / (138.0 \cdot 1.0)] - 1))$
= 0.4527 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4626 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	121.3280	mm
Parallel to Vessel Wall	Rn+tn+t	60.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N42".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9626	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 19.2244	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 19.2244	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.2724	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(19.2244, 3.0)]
= 6.3000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.9626, 6.3)
= 6.3000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.764, tr = 0.453, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.03$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N42"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640	mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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PV Elite 28 U2 Licensee: SL User
FileName : 401-V-32002_ - 7%
Nozzle Calcs.: N4_2"

Noz1: 45 5:15pm Aug 11,2026

Page 130 of 168



HEXAGON

Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8469 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 334.2730 mm

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Input, Nozzle Desc: N1B10"

From: 30

Pressure for Reinforcement Calculations	P	2.3001	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1855.00	mm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		10.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	

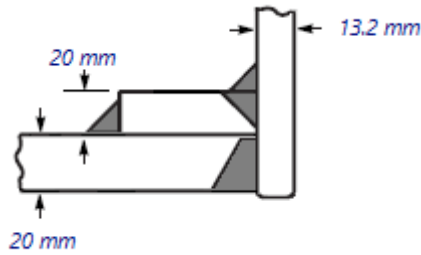
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	233.9240	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	473.0500	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		100.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1B10"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	9.710 in.
Actual Thickness Used in Calculation	0.520 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 17.7034 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 124.82 / (118 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 2.4619 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8117 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	499.2934 mm
Parallel to Vessel Wall, opening length	d	249.6467 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.2500 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

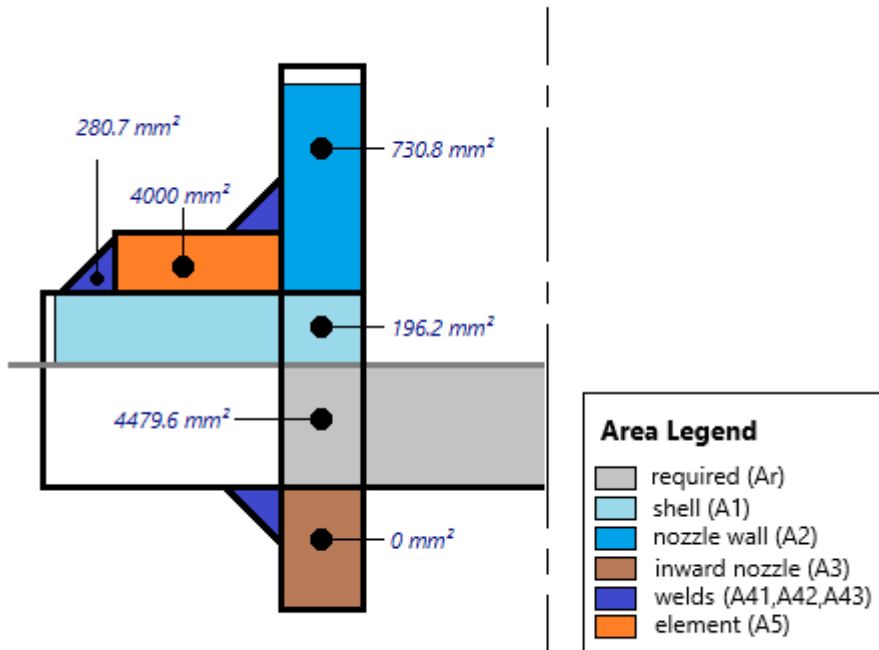


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4479.630	1176.798	NA
Area in Shell	A1	196.172	2265.216	NA
Area in Nozzle Wall	A2	730.821	861.346	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	280.737	280.737	NA
Area in Element	A5	4000.000	4000.000	NA
TOTAL AREA AVAILABLE	Atot	5207.730	7407.299	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 436.6450 20.0000 mm
 Based on given Pad Diameter: 473.0500 16.3595 mm
 Based on the Estimated Diameter Limit: 497.7059 14.5641 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (249.6467 \cdot 17.7034 \cdot 1.0 + 2 \cdot 11.7016 \cdot 17.7034 \cdot 1.0 \cdot (1 - 0.86))$
 $= 4479.630 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 249.647(1.0 \cdot 18.5 - 1.0 \cdot 17.703) - 2 \cdot 11.702$
 $(1.0 \cdot 18.5 - 1.0 \cdot 17.7034) \cdot (1 - 0.855)$
 $= 196.172 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 46.25) \cdot (11.7 - 2.46) \cdot 0.8551$

= 730.821 mm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (100.0) • 0.86 + (0.0) • 0.86 + 195.2286² • 1.0
= 280.737 mm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (473.05 - 273.05) • 20.0 • 1.0
= 4000.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.9619 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 19.2034 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 19.2034 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.2724 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.6026 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[9.603, max(19.2034, 3.0)]
= 9.6026 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9619, 9.6026)
= 9.6026 mm

Available Nozzle Neck Thickness = 13.2016 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.8,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 263.2 MPa	Passed
Occasional	: 11.7,	Allowable	: 156.9 MPa	Passed
Shear	: 14.0,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 20.0, tr = 17.703, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.957$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-10 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.202, tr = 2.462, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.21$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = $2.30/5.10 = 0.451$

Weld Size Calculations, Description: N1B10"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.7016 mm
Intermediate Calc. for pad/shell Welds	TminPad	18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.2500 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (4479.6304 - 196.1716 + 2 \cdot 11.7016 \cdot 0.8551 \cdot (1.0 \cdot 18.5 - 17.7034)) \cdot 138)$
 $= 593256.00 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (730.8213 + 4000.0 + 280.7374 - 0.0 \cdot 0.86) \cdot 138$
 $= 691523.75 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (730.8213 + 0.0 + 85.5088 + (370.2198)) \cdot 138$
 $= 163727.00 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (730.8213 + 0.0 + 280.7374 + 4000.0 + (370.2198)) \cdot 138$
 $= 742608.81 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 273.05 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 247972. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 473.05 \cdot 14.0 \cdot 0.49 \cdot 138$
 $= 703372. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 130.6742) \cdot (13.2016 - 1.5) \cdot 0.7 \cdot 118$
 $= 396762. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 273.05 \cdot 20.0 \cdot 0.74 \cdot 138$
 $= 875923. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 273.05 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 700726. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (703372 + 396762) = 1100134 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (247972 + 875923 + 700726 + 0) = 1824621 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (703372 + 700726 + 0) = 1404098 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 1100134 N	, must exceed W = 593256 N	or W1 = 691523 N
Path 2-2 = 1824621 N	, must exceed W = 593256 N	or W2 = 163727 N
Path 3-3 = 1404098 N	, must exceed W = 593256 N	or W3 = 742608 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 8.9136 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 262.8376 mm

Input Echo, WRC297 Item 1, Description: N1B10" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2100.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	246.647	mm
Nozzle Thickness	13.2016	mm



Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	473.050	mm
Thickness of Reinforcing Pad	20.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	2.30	MPa
Radial Load	P	-6699.98	N
Circumferential Shear	Vc	9999.99	N
Longitudinal Shear	Vl	9999.99	N
Circumferential Moment	Mc	0.750E+07	N-mm
Longitudinal Moment	Ml	0.100E+08	N-mm
Torsional Moment	Mt	0.125E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1070.75(20.0 - 1.5))}$
 $= 737.505 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2141.500	mm
Nozzle Outside Diameter	(d) =	273.050	mm
Vessel Thickness used	(T) =	38.500	mm
Nozzle Thickness used	(t) =	11.702	mm
T / t	=	3.290	
d / t	=	23.334	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.951$$

Nr/P	=	0.092
Mr/P	=	0.101
M0/P	=	0.105
N0/P	=	0.234
MrD/Mc	=	0.165
NrDL/Mc	=	0.130
M0d/Mc	=	0.207
N0DL/Mc	=	0.111
MrD/Ml	=	0.137
NrDL/Ml	=	0.114
M0D/Ml	=	0.165
N0DL/Ml	=	0.272

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-6	-6	6	6
Outplane Bending (Ml)	-24	24	24	-24



Normal Pressure Stress	61	63	61	63
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Outplane Stress Summary	34	79	96	44
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-20	20	20	-20
Inplane Pressure Stress	30	30	30	30

Inplane Stress Summary	10	46	57	10
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	2	2
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	34	80	96	44
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-22	22	22	-22
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30

Outplane Stress Summary	9	49	59	8
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-18	18	18	-18
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	61	63	61	63

Inplane Stress Summary	44	77	85	45
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right



circumferential plane) Outside Inside Outside Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary 2 2 3 3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress 44 77 85 46

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	30	28	30

Hoop Stress Summary 22 25 35 38

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	16	-16	16	-16
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-16	-16	16	16
Axial Bending (ML)	-128	128	128	-128
Axial Pressure Stress	14	14	14	14

Axial Stress Summary -114 111 176 -113

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9

Shear Stress Summary 11 11 7 7

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress 139 112 176 153

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0



Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	30	28	30
Hoop Stress Summary	27	29	31	33

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	16	-16	16	-16
Axial Membrane (Mc)	-12	-12	12	12
Axial Bending (Mc)	-120	120	120	-120
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	14	14	14	14
Axial Stress Summary	-101	106	163	-109

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	9	9	9	9
Shear Stress Summary	7	7	11	11

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	129	107	164	144

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{1060.75(20.0 - 1.5)}$$

$$= 735.331 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	2121.500 mm
Nozzle Outside Diameter (d)	=	473.050 mm
Vessel Thickness used (T)	=	18.500 mm
Nozzle Thickness used (t)	=	111.702 mm
T / t	=	0.166
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.388$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.068
Mr/P	=	0.128
M0/P	=	0.038
N0/P	=	0.063
MrD/Mc	=	0.281
NrDL/Mc	=	0.098
M0d/Mc	=	0.085
N0DL/Mc	=	0.076
MrD/Ml	=	0.098
NrDL/Ml	=	0.087
M0D/Ml	=	0.029
N0DL/Ml	=	0.083



Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	129	131	129	131
Outplane Stress Summary	119	134	151	123

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-36	36	36	-36
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	39	81	122	20

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	119	134	151	123

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64
Outplane Stress Summary	43	81	97	41

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1

Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-78	78	78	-78
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	129	131	129	131

Inplane Stress Summary	63	191	228	44
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	63	191	228	46

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	61	63	61	63	61	63	61	63
Circ. Pl (SUS)	-5	-5	7	7	-2	-2	2	2
Circ. Q (SUS)	-22	22	26	-26	-16	16	20	-20
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	-2	-2	2	2	-1	-1	3	3
Long. Q (SUS)	-18	18	22	-22	-20	20	24	-24
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	61.0	63.0	61.0	63.0	61.0	63.0	61.0	63.0
Pm+Pl (SUS)	56.0	58.0	68.0	70.0	59.0	61.0	63.0	65.0
Pm+Pl+Q (Total)	34.2	80.1	94.1	44.1	43.1	77.1	83.2	45.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.00	138.00	Passed
Pm+Pl (SUS)	70.00	207.00	Passed
Pm+Pl+Q (TOTAL)	94.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of	Stress Values at
---------	------------------

Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	129	131	129	131	129	131	129	131
Circ. Pl (SUS)	-4	-4	6	6	-3	-3	5	5
Circ. Q (SUS)	-6	6	14	-14	-63	63	93	-93
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. Pl (SUS)	-4	-4	6	6	-2	-2	4	4
Long. Q (SUS)	-21	21	51	-51	-19	19	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	129.0	131.0	129.0	131.0	129.0	131.0	129.0	131.0
Pm+Pl (SUS)	125.0	127.0	135.0	137.0	126.0	128.0	134.0	136.0
Pm+Pl+Q (Total)	119.0	133.0	149.0	123.0	63.0	191.0	227.0	43.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+Pl (SUS)	137.00	207.00	Passed
Pm+Pl+Q (TOTAL)	227.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	28	30	28	30	28	30	28	30
Circ. Pl (SUS)	-5	-5	7	7	-1	-1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-16	-16	16	16	-12	-12	12	12
Long. Q (SUS)	-112	112	144	-144	-104	104	136	-136
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	28.0	30.0	28.0	30.0	28.0	30.0	28.0	30.0
Pm+Pl (SUS)	25.1	27.1	35.2	37.1	27.0	29.0	31.2	33.1
Pm+Pl+Q (Total)	138.5	111.2	174.5	151.8	130.0	106.8	162.8	144.4

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30.00	118.00	Passed
Pm+Pl (SUS)	37.14	177.00	Passed
Pm+Pl+Q (TOTAL)	174.46	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 U2 Licensee: SL User
FileName : 401-V-32002_ - 7%
Nozzle Calcs.: N1B_10"

Nozl: 46 5:15pm Aug 11, 2026

Page 144 of 168



HEXAGON

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Input, Nozzle Desc: N1A10"

From: 30

Pressure for Reinforcement Calculations	P	2.3001	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2100.00	mm
Design Length of Section	L	4550.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1855.00	mm
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User Entered Minimum Design Metal Temperature		-33.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		10.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	

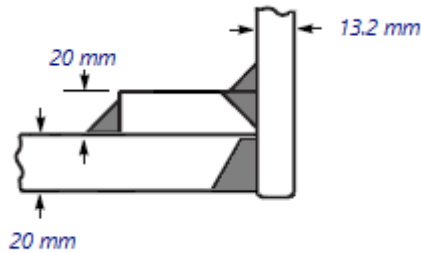
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	233.9240	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	473.0500	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		100.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1A10"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	9.710 in.
Actual Thickness Used in Calculation	0.520 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 1051.5 / (138 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 17.7034 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.3 \cdot 124.82 / (118 \cdot 1.0 - 0.6 \cdot 2.3)$
 $= 2.4619 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8117 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	499.2934 mm
Parallel to Vessel Wall, opening length	d	249.6467 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.2500 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

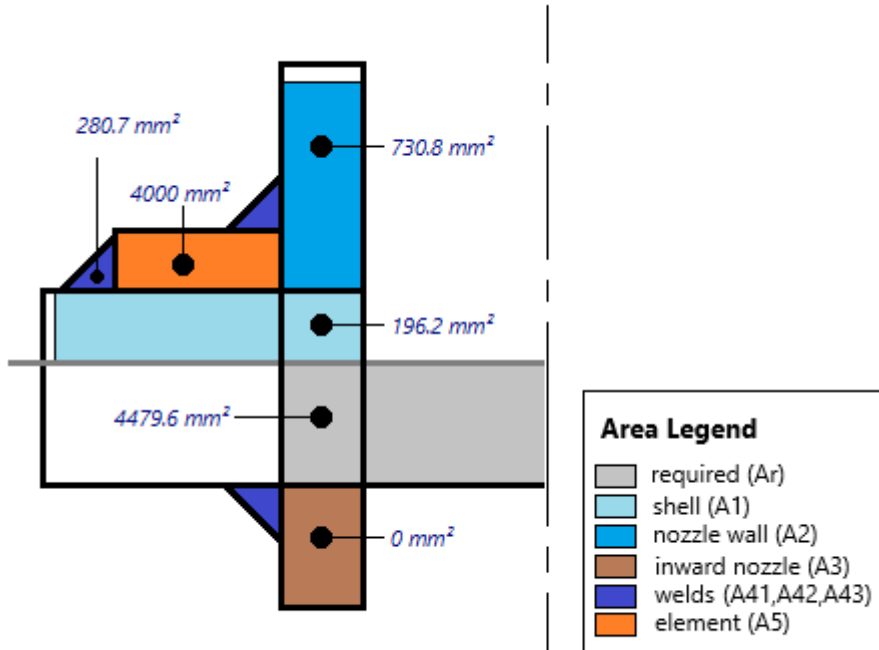


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4479.630	1176.798	NA
Area in Shell	A1	196.172	2265.216	NA
Area in Nozzle Wall	A2	730.821	861.346	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	280.737	280.737	NA
Area in Element	A5	4000.000	4000.000	NA
TOTAL AREA AVAILABLE	Atot	5207.730	7407.299	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 436.6450 20.0000 mm
 Based on given Pad Diameter: 473.0500 16.3595 mm
 Based on the Estimated Diameter Limit: 497.7059 14.5641 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (249.6467 \cdot 17.7034 \cdot 1.0 + 2 \cdot 11.7016 \cdot 17.7034 \cdot 1.0 \cdot (1 - 0.86))$
 $= 4479.630 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 249.647(1.0 \cdot 18.5 - 1.0 \cdot 17.703) - 2 \cdot 11.702$
 $(1.0 \cdot 18.5 - 1.0 \cdot 17.7034) \cdot (1 - 0.855)$
 $= 196.172 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 46.25) \cdot (11.7 - 2.46) \cdot 0.8551$



= 730.821 mm²

Area Available in Welds [A41 + A42 + A43]:

= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (100.0) • 0.86 + (0.0) • 0.86 + 195.2286² • 1.0
= 280.737 mm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (473.05 - 273.05) • 20.0 • 1.0
= 4000.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.9619 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 19.2034 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 19.2034 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.2724 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.6026 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[9.603, max(19.2034, 3.0)]
= 9.6026 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(3.9619, 9.6026)
= 9.6026 mm

Available Nozzle Neck Thickness = 13.2016 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.8,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 263.2 MPa	Passed
Occasional	: 11.7,	Allowable	: 156.9 MPa	Passed
Shear	: 14.0,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 20.0, tr = 17.703, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.957$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -10 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.202, tr = 2.462, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.21$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $2.30/5.10 = 0.451$

Weld Size Calculations, Description: N1A10"

Intermediate Calc. for nozzle/shell Welds Tmin 11.7016 mm
 Intermediate Calc. for pad/shell Welds TminPad 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.2500 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (4479.6304 - 196.1716 + 2 \cdot 11.7016 \cdot 0.8551 \cdot (1.0 \cdot 18.5 - 17.7034)) \cdot 138)$
 $= 593256.00 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (730.8213 + 4000.0 + 280.7374 - 0.0 \cdot 0.86) \cdot 138$
 $= 691523.75 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (730.8213 + 0.0 + 85.5088 + (370.2198)) \cdot 138$
 $= 163727.00 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (730.8213 + 0.0 + 280.7374 + 4000.0 + (370.2198)) \cdot 138$
 $= 742608.81 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 273.05 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 247972. \text{ N}$



Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 473.05 \cdot 14.0 \cdot 0.49 \cdot 138$
 $= 703372. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 130.6742) \cdot (13.2016 - 1.5) \cdot 0.7 \cdot 118$
 $= 396762. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 273.05 \cdot 20.0 \cdot 0.74 \cdot 138$
 $= 875923. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 273.05 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 700726. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (703372 + 396762) = 1100134 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (247972 + 875923 + 700726 + 0) = 1824621 \text{ N}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (703372 + 700726 + 0) = 1404098 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 1100134 N	, must exceed W = 593256 N	or W1 = 691523 N
Path 2-2 = 1824621 N	, must exceed W = 593256 N	or W2 = 163727 N
Path 3-3 = 1404098 N	, must exceed W = 593256 N	or W3 = 742608 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 8.9136 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 262.8376 mm

Input Echo, WRC297 Item 1, Description: N1A10" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2100.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	246.647	mm
Nozzle Thickness	13.2016	mm



Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	473.050	mm
Thickness of Reinforcing Pad	20.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	2.30	MPa
Radial Load	P	-6699.98	N
Circumferential Shear	Vc	9999.99	N
Longitudinal Shear	Vl	9999.99	N
Circumferential Moment	Mc	0.750E+07	N-mm
Longitudinal Moment	Ml	0.100E+08	N-mm
Torsional Moment	Mt	0.125E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1070.75(20.0 - 1.5))}$
 $= 737.505 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2141.500	mm
Nozzle Outside Diameter	(d) =	273.050	mm
Vessel Thickness used	(T) =	38.500	mm
Nozzle Thickness used	(t) =	11.702	mm
T / t	=	3.290	
d / t	=	23.334	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.951$$

Nr/P	=	0.092
Mr/P	=	0.101
M0/P	=	0.105
N0/P	=	0.234
MrD/Mc	=	0.165
NrDL/Mc	=	0.130
M0d/Mc	=	0.207
N0DL/Mc	=	0.111
MrD/Ml	=	0.137
NrDL/Ml	=	0.114
M0D/Ml	=	0.165
N0DL/Ml	=	0.272

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-6	-6	6	6
Outplane Bending (Ml)	-24	24	24	-24



Normal Pressure Stress	61	63	61	63
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Outplane Stress Summary	34	79	96	44
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Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-20	20	20	-20
Inplane Pressure Stress	30	30	30	30

Inplane Stress Summary	10	46	57	10
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Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	2	2
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Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	34	80	96	44
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-22	22	22	-22
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30

Outplane Stress Summary	9	49	59	8
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-18	18	18	-18
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	61	63	61	63

Inplane Stress Summary	44	77	85	45
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to	Cu Left	Cl Left	Du Right	Dl Right
--	------------	------------	-------------	-------------



circumferential plane) Outside Inside Outside Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary 2 2 3 3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress 44 77 85 46

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	30	28	30

Hoop Stress Summary 22 25 35 38

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	16	-16	16	-16
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-16	-16	16	16
Axial Bending (ML)	-128	128	128	-128
Axial Pressure Stress	14	14	14	14

Axial Stress Summary -114 111 176 -113

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9

Shear Stress Summary 11 11 7 7

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress 139 112 176 153

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0



Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	30	28	30
Hoop Stress Summary	27	29	31	33

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	16	-16	16	-16
Axial Membrane (Mc)	-12	-12	12	12
Axial Bending (Mc)	-120	120	120	-120
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	14	14	14	14
Axial Stress Summary	-101	106	163	-109

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	9	9	9	9
Shear Stress Summary	7	7	11	11

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	129	107	164	144

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{1060.75(20.0 - 1.5)}$$

$$= 735.331 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2121.500 mm
Nozzle Outside Diameter	(d) =	473.050 mm
Vessel Thickness used	(T) =	18.500 mm
Nozzle Thickness used	(t) =	111.702 mm
T / t	=	0.166
d / T	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 2.388$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.068
Mr/P	=	0.128
M0/P	=	0.038
N0/P	=	0.063
MrD/Mc	=	0.281
NrDL/Mc	=	0.098
M0d/Mc	=	0.085
N0DL/Mc	=	0.076
MrD/Ml	=	0.098
NrDL/Ml	=	0.087
M0D/Ml	=	0.029
N0DL/Ml	=	0.083



Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	129	131	129	131
Outplane Stress Summary	119	134	151	123

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-36	36	36	-36
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	39	81	122	20

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	119	134	151	123

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64
Outplane Stress Summary	43	81	97	41

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1

Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-78	78	78	-78
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	129	131	129	131

Inplane Stress Summary	63	191	228	44
------------------------	----	-----	-----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	63	191	228	46

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	61	63	61	63	61	63	61	63
Circ. Pl (SUS)	-5	-5	7	7	-2	-2	2	2
Circ. Q (SUS)	-22	22	26	-26	-16	16	20	-20
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	-2	-2	2	2	-1	-1	3	3
Long. Q (SUS)	-18	18	22	-22	-20	20	24	-24
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	61.0	63.0	61.0	63.0	61.0	63.0	61.0	63.0
Pm+Pl (SUS)	56.0	58.0	68.0	70.0	59.0	61.0	63.0	65.0
Pm+Pl+Q (Total)	34.2	80.1	94.1	44.1	43.1	77.1	83.2	45.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.00	138.00	Passed
Pm+Pl (SUS)	70.00	207.00	Passed
Pm+Pl+Q (TOTAL)	94.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of	Stress Values at
---------	------------------

Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	129	131	129	131	129	131	129	131
Circ. Pl (SUS)	-4	-4	6	6	-3	-3	5	5
Circ. Q (SUS)	-6	6	14	-14	-63	63	93	-93
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. Pl (SUS)	-4	-4	6	6	-2	-2	4	4
Long. Q (SUS)	-21	21	51	-51	-19	19	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	129.0	131.0	129.0	131.0	129.0	131.0	129.0	131.0
Pm+Pl (SUS)	125.0	127.0	135.0	137.0	126.0	128.0	134.0	136.0
Pm+Pl+Q (Total)	119.0	133.0	149.0	123.0	63.0	191.0	227.0	43.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+Pl (SUS)	137.00	207.00	Passed
Pm+Pl+Q (TOTAL)	227.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	28	30	28	30	28	30	28	30
Circ. Pl (SUS)	-5	-5	7	7	-1	-1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-16	-16	16	16	-12	-12	12	12
Long. Q (SUS)	-112	112	144	-144	-104	104	136	-136
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	28.0	30.0	28.0	30.0	28.0	30.0	28.0	30.0
Pm+Pl (SUS)	25.1	27.1	35.2	37.1	27.0	29.0	31.2	33.1
Pm+Pl+Q (Total)	138.5	111.2	174.5	151.8	130.0	106.8	162.8	144.4

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30.00	118.00	Passed
Pm+Pl (SUS)	37.14	177.00	Passed
Pm+Pl+Q (TOTAL)	174.46	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 U2 Licensee: SL User
FileName : 401-V-32002_ - 7%
Nozzle Calcs.: N1A_10"

Nozl: 47 5:15pm Aug 11, 2026

Page 158 of 168



HEXAGON

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Input, Nozzle Desc: N214"

From: 50

Pressure for Reinforcement Calculations	P	2.3001	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	2100.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	20.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		14.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	

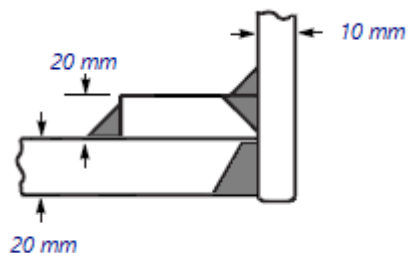
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	322.2240	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	15.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	615.6000	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		120.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



```

= min( fr2, fr4 )
= min( 1.0, 1.0 )
= 1.000

```

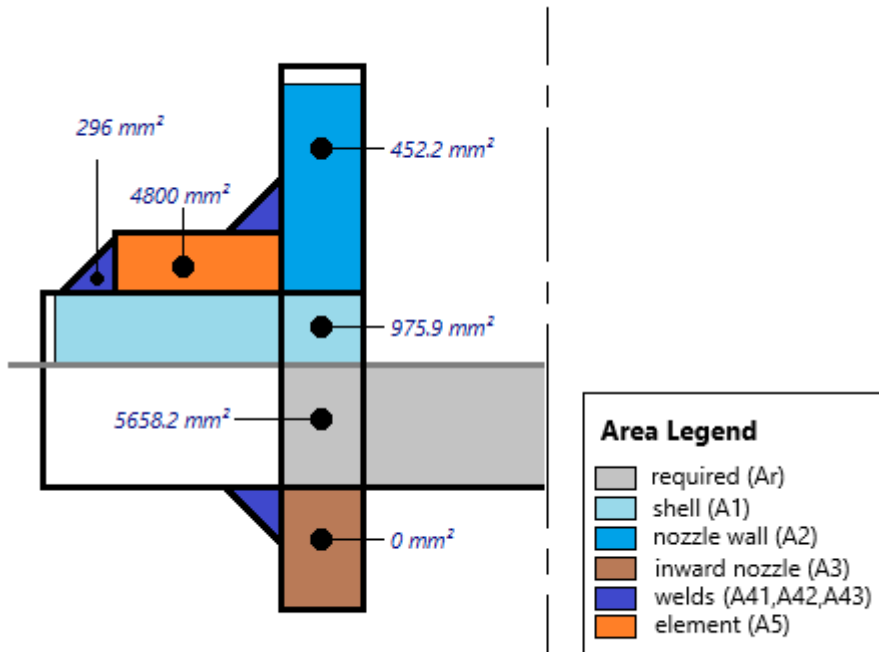


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5658.183	983.347	NA
Area in Shell	A1	975.917	4667.406	NA
Area in Nozzle Wall	A2	452.208	609.078	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	296.000	296.000	NA
Area in Element	A5	4800.000	4800.000	NA
TOTAL AREA AVAILABLE	Atot	6524.125	10372.484	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	572.3029	20.0000 mm
Based on given Pad Diameter:	615.6000	16.3919 mm
Based on the Estimated Diameter Limit:	715.6125	11.5703 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (358.6 \cdot 15.7785 \cdot 1.0 + 2 \cdot 8.5 \cdot 15.7785 \cdot 1.0 \cdot (1 - 1.0))$
 $= 5658.183 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 358.6(1.0 \cdot 18.5 - 1.0 \cdot 15.779) - 2 \cdot 8.5$
 $(1.0 \cdot 18.5 - 1.0 \cdot 15.7785) \cdot (1 - 1.0)$
 $= 975.917 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 41.25) \cdot (8.5 - 3.02) \cdot 1.0$



= 452.208 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0$
= 296.000 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(615.6 - 375.6) \cdot 20.0 \cdot 1.0$
= 4800.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.5187 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 19.0223 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 19.0223 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.2707 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.8312 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[9.831, max(19.0223, 3.0)]
= 9.8312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(4.5187, 9.8312)
= 9.8312 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 50.6,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 294.4 MPa	Passed
Occasional	: 23.7,	Allowable	: 183.5 MPa	Passed
Shear	: 18.4,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 20.0, tr = 15.779, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.853$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-16 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 3.019, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.355$, Temp. Reduction = 74 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $2.30/5.10 = 0.451$

Weld Size Calculations, Description: N214"

Intermediate Calc. for nozzle/shell Welds Tmin 8.5000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$9.8980 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (5658.1826 - 975.9174 + 2 \cdot 8.5 \cdot 1.0 \cdot (1.0 \cdot 18.5 - 15.7785)) \cdot 138)$
 $= 652469.81 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (452.2076 + 4800.0 + 296.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 765573.69 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (452.2076 + 0.0 + 100.0 + (314.5)) \cdot 138$
 $= 119593.30 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (452.2076 + 0.0 + 296.0 + 4800.0 + (314.5)) \cdot 138$
 $= 808970.25 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 375.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 398918. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 615.6 \cdot 14.0 \cdot 0.49 \cdot 138$
 $= 915328. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 183.55) \cdot (10.0 - 1.5) \cdot 0.7 \cdot 138$



= 473438. N

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 375.6 \cdot 20.0 \cdot 0.74 \cdot 138$$

$$= 1204895. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 375.6 \cdot (15.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 903672. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (915328 + 473438) = 1388767 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (398918 + 1204895 + 903672 + 0) = 2507485 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (915328 + 903672 + 0) = 1819000 N$$

Summary of Failure Path Calculations:

Path 1-1 = 1388766 N	, must exceed W = 652469 N	or W1 = 765573 N
Path 2-2 = 2507485 N	, must exceed W = 652469 N	or W2 = 119593 N
Path 3-3 = 1818999 N	, must exceed W = 652469 N	or W3 = 808970 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 9.2746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 351.4986 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.735 %

Input Echo, WRC107/537 Item 1, Description: N214" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	3780.000	mm
Vessel Thickness	Tv	20.000	mm
Design Temperature	T1	80.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow

Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	355.600	mm
Nozzle Thickness	Tn	10.000	mm
Nozzle Material		SA-516 70	



Nozzle UNS Number		K02700	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Thickness of Reinforcing Pad	Tpad	20.000	mm
Diameter of Reinforcing Pad	Dpad	615.600	mm
Design Internal Pressure	Dp	2.300	MPa
Include Pressure Thrust		No	

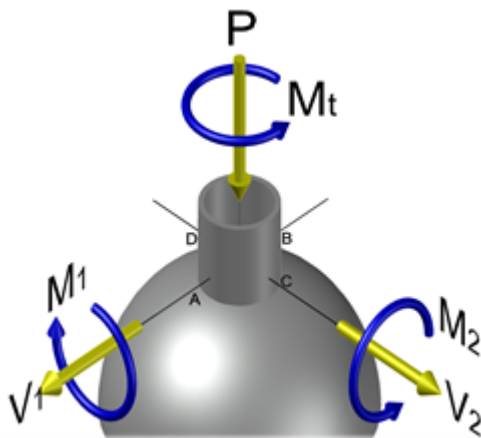
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-9400.0	N
Longitudinal Shear (SUS)	(V1) V1	14000.0	N
Circumferential Shear (SUS)	(Vc) V2	14000.0	N
Circumferential Moment (SUS)	(Mc) M1	14699987.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	19600028.0	N-mm
Torsional Moment (SUS)	Mt	24499934.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 375.6 + 2 \cdot 1.65 \cdot \sqrt{(1900.75 (20.0 - 1.5))} \\
 &= 994.417 \text{ mm}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-9400.0	N
Circumferential Shear	(VC) V2	14000.0	N
Longitudinal Shear	(VL) V1	14000.0	N
Circumferential Moment	(MC) M1	14699987.0	N-mm
Longitudinal Moment	(ML) M2	19600028.0	N-mm
Torsional Moment	MT	24499934.0	N-mm

Dimensionless Parameters: U = 0.69 TAU = 21.59 RHO = 4.53

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
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N(x) * T / P	SP 7	0.02801	(A,B,C,D)
M(x) / P	SP 7	0.01332	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 7	0.06287	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 7	0.03060	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 7	0.06287	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 7	0.03060	(A,B,C,D)
N(y) * T / P	SP 7	0.27050	(A,B,C,D)
M(y) / P	SP 7	0.04901	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 7	0.35621	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 7	0.13344	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 7	0.35621	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 7	0.13344	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad. Bend. P		0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5
Rad. Memb. MC		0.0	0.0	0.0	0.0	-2.3	-2.3	2.3	2.3
Rad. Bend. MC		0.0	0.0	0.0	0.0	-6.7	6.7	6.7	-6.7
Rad. Memb. ML		-3.1	-3.1	3.1	3.1	0.0	0.0	0.0	0.0
Rad. Bend. ML		-8.9	8.9	8.9	-8.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-11.3	5.6	12.7	-6.2	-8.3	4.1	9.7	-4.7
Tang. Memb. P		1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Tang. Bend. P		1.9	-1.9	1.9	-1.9	1.9	-1.9	1.9	-1.9
Tang. Memb. MC		0.0	0.0	0.0	0.0	-13.0	-13.0	13.0	13.0
Tang. Bend. MC		0.0	0.0	0.0	0.0	-29.3	29.3	29.3	-29.3
Tang. Memb. ML		-17.4	-17.4	17.4	17.4	0.0	0.0	0.0	0.0
Tang. Bend. ML		-39.0	39.0	39.0	-39.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-52.8	21.5	60.0	-21.8	-38.7	16.1	45.9	-16.4
Shear VC		0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9
Tot. Shear		3.5	3.5	2.3	2.3	2.3	2.3	3.5	3.5
Str. Int.		53.1	22.2	60.1	22.1	38.9	16.5	46.2	17.4

Unitless Prm: U = 1.64 TAU = 0.00 (35.71) RHO = 0.00 (2.18)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.03341	(A,B,C,D)
M(x) / P	SR 2	0.01616	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.03200	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.02255	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.03200	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.02255	(A,B,C,D)
N(y) * T / P	SR 2	0.00995	(A,B,C,D)
M(y) / P	SR 2	0.00484	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.00959	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.00667	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.00959	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.00667	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00



Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Rad. Bend. P		2.7	-2.7	2.7	-2.7	2.7	-2.7	2.7	-2.7
Rad. Memb. MC		0.0	0.0	0.0	0.0	-7.3	-7.3	7.3	7.3
Rad. Bend. MC		0.0	0.0	0.0	0.0	-31.0	31.0	31.0	-31.0
Rad. Memb. ML		-9.8	-9.8	9.8	9.8	0.0	0.0	0.0	0.0
Rad. Bend. ML		-41.3	41.3	41.3	-41.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-47.5	29.8	54.7	-33.3	-34.7	21.9	41.9	-25.4
Tang. Memb. P		0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Tang. Bend. P		0.8	-0.8	0.8	-0.8	0.8	-0.8	0.8	-0.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Tang. Bend. MC		0.0	0.0	0.0	0.0	-9.2	9.2	9.2	-9.2
Tang. Memb. ML		-2.9	-2.9	2.9	2.9	0.0	0.0	0.0	0.0
Tang. Bend. ML		-12.2	12.2	12.2	-12.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-14.1	8.8	16.2	-9.8	-10.3	6.4	12.4	-7.5
Shear VC		0.8	0.8	-0.8	-0.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT		2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Tot. Shear		3.0	3.0	1.4	1.4	1.4	1.4	3.0	3.0
Str. Int.		47.8	30.2	54.7	33.4	34.8	22.0	42.2	25.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		56.5	56.5	56.5	56.5	56.5	56.5	56.5	56.5
Rad. Pl (SUS)		-2.9	-2.9	3.2	3.2	-2.1	-2.1	2.5	2.5
Rad. Q (SUS)		-8.4	8.4	9.5	-9.5	-6.2	6.2	7.2	-7.2
Long. Pm (SUS)		56.5	56.5	56.5	56.5	56.5	56.5	56.5	56.5
Long. Pl (SUS)		-15.6	-15.6	19.1	19.1	-11.3	-11.3	14.7	14.7
Long. Q (SUS)		-37.2	37.2	40.9	-40.9	-27.4	27.4	31.1	-31.1
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear Q (SUS)		2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9
Pm (SUS)		56.5	56.5	56.5	56.5	56.5	56.5	56.5	56.5
Pm+Pl (SUS)		53.7	53.7	75.6	75.6	54.4	54.4	71.3	71.3
Pm+Pl+Q (Total)		45.5	78.8	116.6	50.6	48.4	73.0	102.7	52.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	56.52	138.00	Passed
Pm+Pl (SUS)	75.62	207.00	Passed
Pm+Pl+Q (TOTAL)	116.59	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	117.6	117.6	117.6	117.6	117.6	117.6	117.6	117.6
Rad.	Pl (SUS)	-8.9	-8.9	10.7	10.7	-6.4	-6.4	8.2	8.2
Rad.	Q (SUS)	-38.6	38.6	44.0	-44.0	-28.3	28.3	33.6	-33.6
Long.	Pm (SUS)	117.6	117.6	117.6	117.6	117.6	117.6	117.6	117.6
Long.	Pl (SUS)	-2.7	-2.7	3.2	3.2	-1.9	-1.9	2.5	2.5
Long.	Q (SUS)	-11.4	11.4	13.0	-13.0	-8.4	8.4	10.0	-10.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.8	0.8	-0.8	-0.8	-0.8	-0.8	0.8	0.8
Shear	Q (SUS)	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Pm	(SUS)	117.6	117.6	117.6	117.6	117.6	117.6	117.6	117.6
Pm+Pl	(SUS)	115.0	115.0	128.4	128.4	115.8	115.8	125.9	125.9
Pm+Pl+Q	(Total)	103.8	147.8	172.3	107.9	107.4	139.6	159.8	110.6

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	117.59	138.00	Passed
Pm+Pl (SUS)	128.36	207.00	Passed
Pm+Pl+Q (TOTAL)	172.30	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: